Active Inference Multi-Track Exemplar

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How to verify

- Scan **Integrity** QR and compare the embedded SHA-256 prefix to the table above.
- Scan **Zenodo** / **GitHub** QR codes and confirm they resolve to this release pairing.
- Full hashes and structured fields: `../data/transmission_manifest.json`.



Figure 1: Integrity QR strip

Structured manifest: `../data/transmission_manifest.json`
Stego: off | overlays text | barcodes on | XMP on | manifest on → `./secure_run.sh`

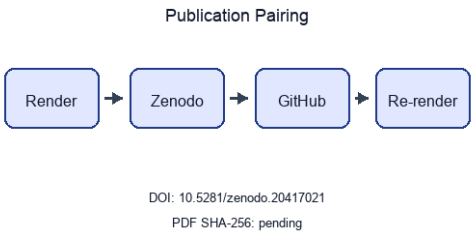


Figure 2: Publication pairing flow

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1 Sheaf Track Coverage

This page summarizes which **sheaf fragment tracks** are bound for each IMRAD row in `manuscript/sheaf/manifest.yaml`. The matrix is regenerated at compose time.

Totals: 95 present / 95 bound / 0 missing (gray).

Color	Meaning
Black	Track **present** (bound and fragment exists)
White	**Absent** (not bound for this row)
Gray	**Missing** (bound but fragment file absent)

1.1 Introduction

- **Introduction** *(group)*
- **Motivation and scope**
- **Contributions**

1.2 Methods

- **Methods** *(group)*
- **Bernoulli–Ising analytical model**
- **pymdp simulation harness**
- **Lean formalization boundary**
- **Sheaf composition**

1.3 Results

- **Results** *(group)*
- **Mutual-information parameter sweep**
- **Free-energy decomposition**
- **T-maze active-inference rollout**
- **Validation invariants**

1.4 Discussion

- **Discussion** *(group)*
- **Limitations and outlook**

1.5 Appendix

- **Appendix** *(group)*
- **Appendix: full track coverage**

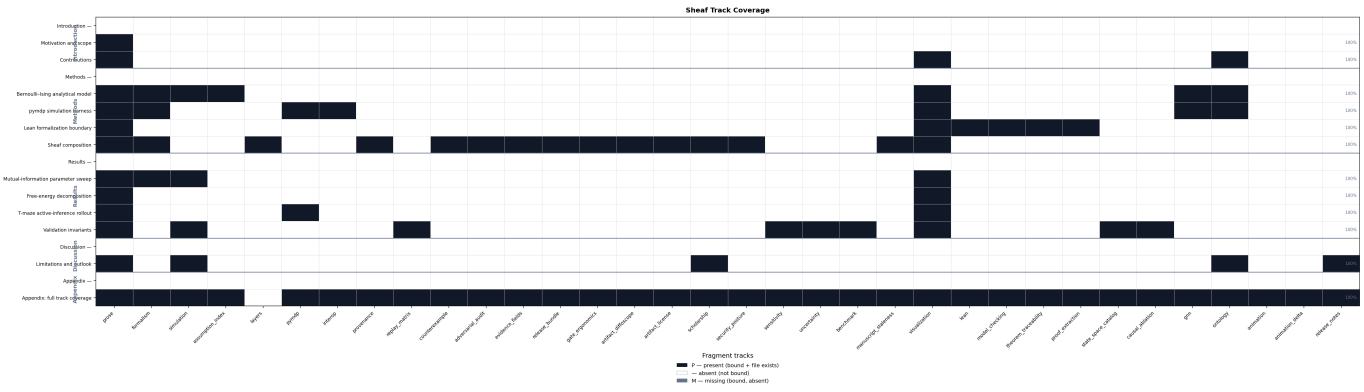


Figure 3: Sheaf track coverage matrix: 17 IMRAD rows × 34 fragment columns. Black = present (P), white = absent (—), gray = missing (M). Counts: 95 present / 95 bound / 0 missing. Generated from `output/data/sheaf_coverage_matrix.json`.

Appendix row `16_appendix_full_sheaf.md` binds 33 fragment track types as a composability proof (registry defines 34 types; optional `layers` is methods-only).

2 Abstract

We study a minimal Active Inference stack on toy models: a Bernoulli–Ising analytical oracle, a pymdp T-maze rollout, and a sheaf-indexed compose contract that binds 34 fragment tracks into 12 flat IMRAD sections. The methodological contribution is a discipline rather than a domain finding: every reported number is hydrated from a generated artifact and every cross-track claim is machine-checked before rendering, so no figure or statistic can drift from the artifact that produced it — 6 sheaf axioms are verified before composition and 25 negative controls keep each failure path live. Claims are limited to those models and their generated artifacts.

sec. 1 reports a 17-row coverage matrix (5 IMRAD group headers) regenerated from the live manifest at compose time. sec. 6 documents the T-maze harness aligned with pymdp sophisticated_inference examples.

sec. 12 records 12 / 12 invariant checks passed. SI planning horizon: 2 steps. Sweep RMSE 0 nats bounds analytical–empirical agreement on the coupling grid.

Introduction

3 Motivation and scope

3.1 Scientific scope

This manuscript couples three tracks on toy Active Inference models: a Bernoulli–Ising analytical oracle, a pymdp T-maze rollout, and a sheaf-indexed assembly contract that binds 34 optional fragment tracks under an IMRAD outline. The conceptual lineage is the free-energy and active-inference literature [Friston, 2010, Buckley et al., 2017, Parr et al., 2022], with critical scope pressure from accounts that separate FEP's broad organizing role from direct empirical brain claims [Gershman, 2019]. Here that distinction is operational: the scientific claims stay within these models and their generated artifacts, not biological agents.

3.2 Manuscript structure

Three **scientific tracks** (analytical, pymdp, sheaf composition) map onto 34 **composable fragment types** and 31 pipeline gates (fig. 4). sec. 1 summarizes which fragment tracks bind to each manifest row. sec. 8 documents the compose pipeline, coverage semantics (eq. 3), and strict validation gates.

The pymdp track follows the pymdp sophisticated_inference examples [Heins et al., 2022] with a minimal T-maze and planning horizon `policy_len = 2`. Other sections cite sec. 6 instead of repeating that reference.

4 Contributions

4.1 Scientific contributions

1. **Analytical oracle** (sec. 5): closed-form mutual information and free-energy decomposition on a symmetric Bernoulli–Ising toy with an independent exact-recomputation cross-check (sec. 9, sec. 10).
2. **Active-inference harness** (sec. 6): deterministic pymdp T-maze rollout — default `state_inference` belief filtering, with sophisticated expected-free-energy policy inference selectable via `mode: policy_inference` — with logged beliefs, actions, and merged invariant gates (sec. 11, sec. 12).
3. **Sheaf-indexed composition** (sec. 8): 34 optional fragment types bind to 17 manifest rows under eq. 3, with a 33-track appendix composability proof (sec. 14).

fig. 4 maps the three scientific tracks to 31 pipeline gates and 34 composable fragment renderers. Measured invariant checks: 12 / 12 passed.

Ontology-facing symbols are checked per model: the Bernoulli toy binds `pi1`, `pi2`, `J`, `gamma`, and `q_joint`, while the SI T-maze binds `location`, `observation`, `policy`, and `belief_entropy` to **HiddenState**, **ObservationLikelihood**, **PolicyPosterior**, and **BeliefEntropy** (fig. 6, sec. 6).

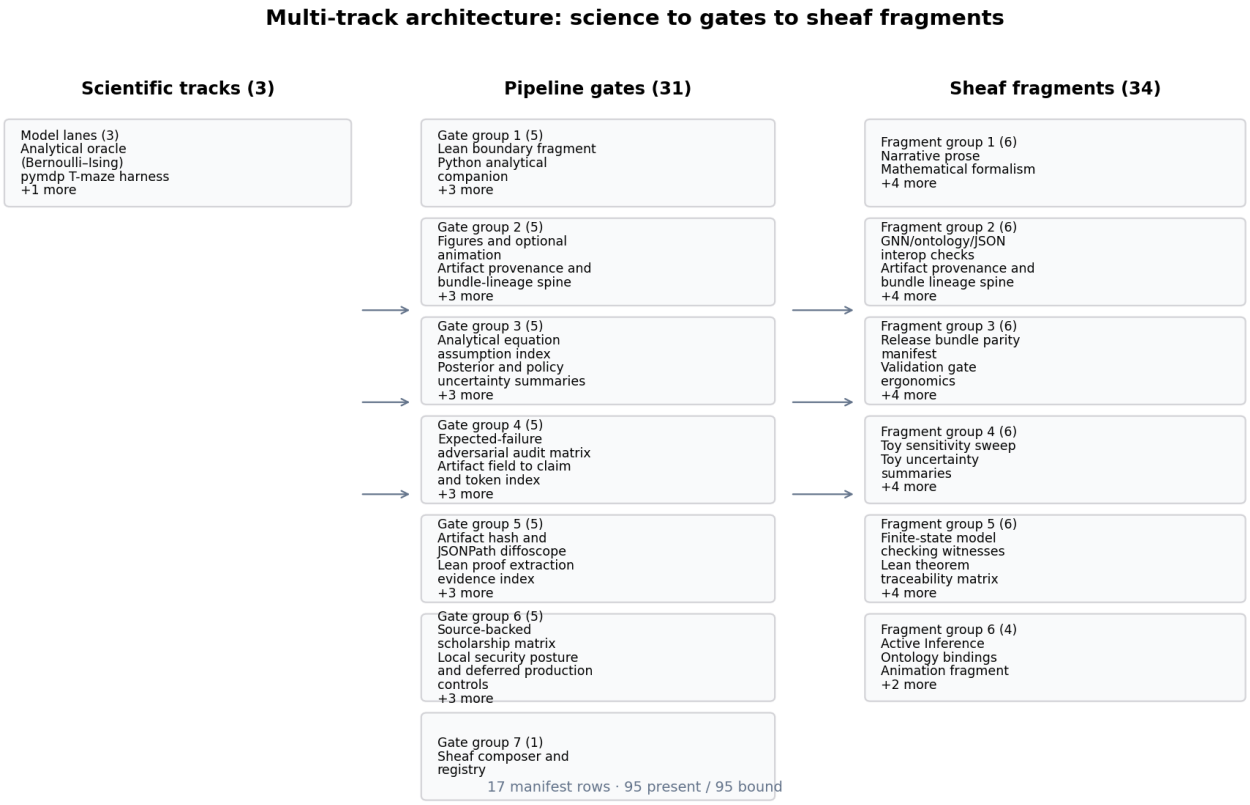


Figure 4: Multi-track architecture: analytical, pymdp, and sheaf composition lanes mapped to 31 pipeline gates and 34 composable fragment types.

4.1.1 Ontology bindings

- `expected_free_energy` → **ExpectedFreeEnergy**
- `location` → **HiddenState**
- `observation` → **ObservationLikelihood**
- `policy` → **PolicyPosterior**

Methods

5 Bernoulli–Ising analytical model

The analytical method is a finite **K=2 Bernoulli / Ising** oracle. The entangled joint eq. 1 gives closed-form mutual information $I(\lambda)$; `output/data/parameter_sweep.csv` then checks the same curve by an independent exact total-correlation recomputation before the value is used in sec. 9. GNN and ontology rows share the same symbol surface (fig. 6), so the derivation, sweep, and model notation are one audited toy contract rather than parallel descriptions.

The scope is intentionally small: finite variational quantities only, no sampling, and no empirical generalization. "Free energy" here means exactly computed variational functionals on this tiny discrete state-space, aligned with mathematical FEP reviews [Buckley et al., 2017], not continuous-time or biological FEP dynamics [Gershman, 2019]. The generated sweep contains 21 grid points, and the merged invariant report records 12 / 12 passing checks.

The entangled joint over binary policies satisfies

$$q_\lambda(\pi) \propto E(\pi)\, \exp(\lambda J(\pi)), \tag{1}$$

with symmetric Ising coupling J and deformation parameter λ. Let $\sigma(\lambda) = q_\lambda(\pi_1 = \pi_2)$ be the probability that the two streams agree (the diagonal mass of the 2×2 joint); by symmetry both marginals are uniform. With binary entropy $H_b(p) = -p\log p - (1-p)\log(1-p)$ in nats, the joint entropy is $H(q_\lambda) = \log 2 + H_b(\sigma(\lambda))$ while each marginal contributes $\log 2$, so the mutual information is

$$I(\lambda) = \sum_k H(q_k) - H(q_\lambda) = \log 2 - H_b(\sigma(\lambda)),$$

vanishing at $\lambda = 0$ ($\sigma = \frac{1}{2}$, independent streams) and saturating at $\log 2$ as $\lambda \to \infty$ ($\sigma \to 1$, perfectly entangled). These symbols are the rows of `analytical_assumption_index.json`, so the derivation is auditable rather than asserted.

The analytical track writes a parameter sweep comparing closed-form mutual information with an independent exact recomputation of it (via total correlation) across $\lambda \in [0, 4]$ on 21 grid points (sec. 9, fig. 5).

The `assumption_index` fragment makes the analytical equations inspectable as a generated artifact instead of relying on prose labels. `output/data/analytical_assumption_index.json` indexes 7 finite-model equation identifiers and 7 rows; the hydrated pass flag is `true`.

The index is deliberately narrow. It covers the Bernoulli-Ising toy equations, their finite binary state assumptions, and the generated artifacts that test the same symbols. Any missing equation identifier or empty assumption list fails the toy-sweep validation gate.

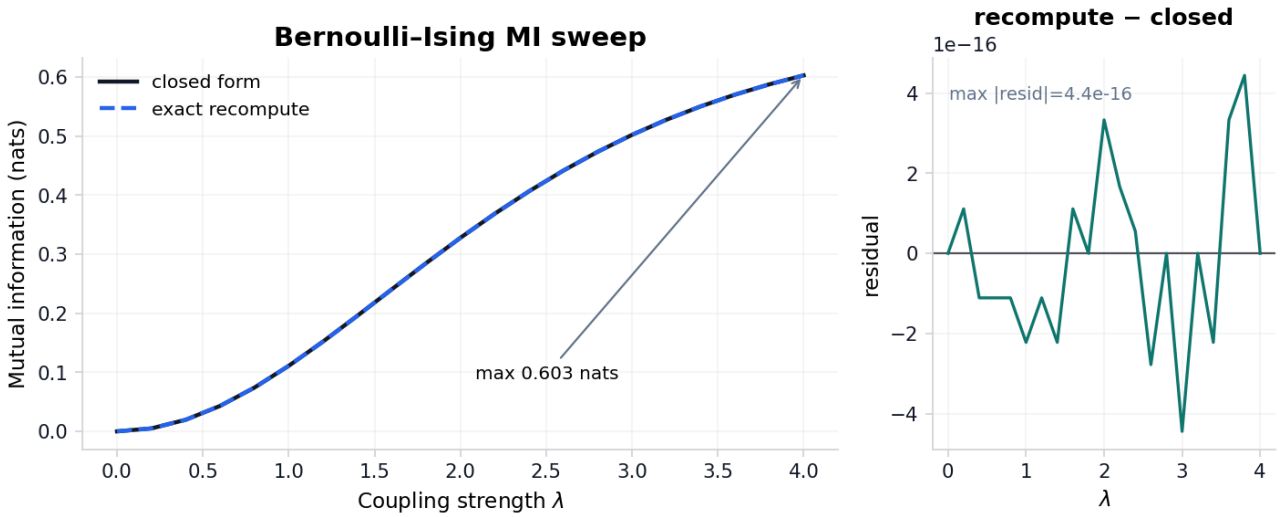


Figure 5: Closed-form $I(\lambda)$ and an independent exact recomputation via total correlation for the symmetric Bernoulli-Ising toy across 21 grid points up to $\lambda_{\max} = 4$; grid maximum 0.6031 nats. Both estimators are deterministic (no sampling), so the right panel is a cross-implementation agreement check (max residual 0 nats), not a sampling residual.

The Bernoulli toy is declared in `gnn/bernoulli_toy.gnn.md` (GNN v1.1), following the GNN notation role described by Smekal and Friedman [Smékal and Friedman, 2023]. fig. 6 links GNN variables to Active Inference Ontology terms bound in the analytical ontology fragment; round-trip parity is checked before render.

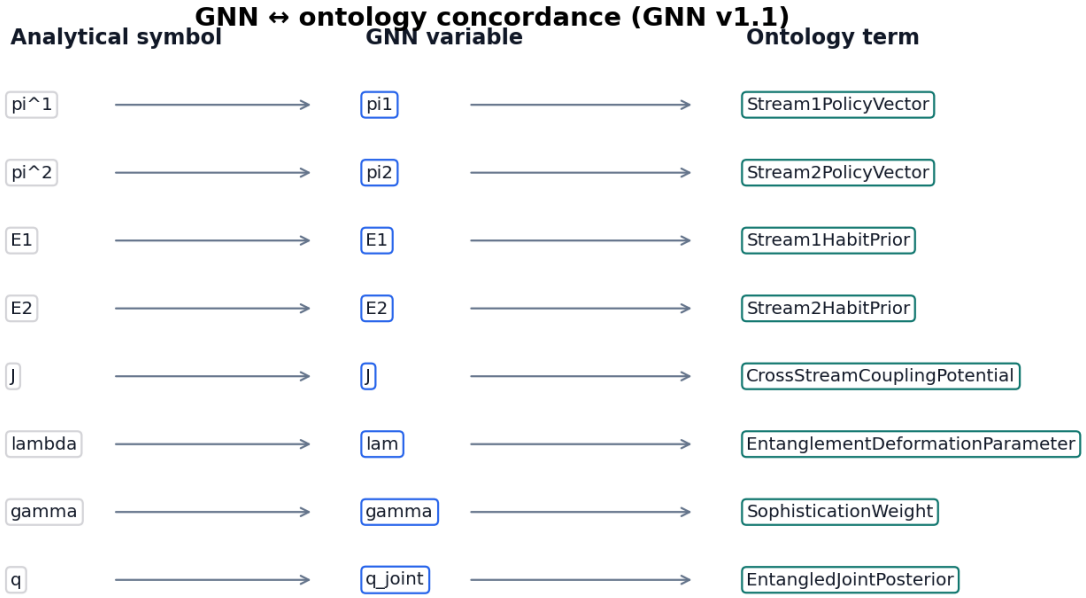


Figure 6: GNN ↔ ontology concordance for the Bernoulli–Ising toy (GNN v1.1).

Measured MI and sweep artifacts in sec. 9 ground the same symbol map used in the concordance diagram.

5.0.1 Ontology bindings

- `E1` → **Stream1HabitPrior**
- `E2` → **Stream2HabitPrior**
- `J` → **CrossStreamCouplingPotential**
- `gamma` → **SophisticationWeight**
- `lam` → **EntanglementDeformationParameter**
- `pi1` → **Stream1PolicyVector**
- `pi2` → **Stream2PolicyVector**
- `q_joint` → **EntangledJointPosterior**

6 pymdp simulation harness

Sophisticated inference (planning horizon). The pymdp method is a deterministic state-inference harness on a minimal T-maze (fig. 7) with planning horizon `policy_len = 2`. The discrete-state framing follows finite POMDP active-inference treatments and sophisticated-inference analyses [Da Costa et al., 2020, Smith et al., 2022, Friston et al., 2021, Da Costa et al., 2023], and the implementation anchor is the pymdp software paper [Heins et al., 2022]. The default `state_inference` rollout writes the summary/trace artifacts used in sec. 11; mean belief entropy is 0.0000.

The method keeps runtime, posterior, and extension evidence separate. `output/data/si_policy_comparison.json` compares `state_inference` and `policy_inference` over declared toy horizons and seeds without replacing the default rollout (4 rows; complete-grid flag 1). Agent construction and backend warnings live in `output/reports/pymdp_runtime_diagnostics.json` (4 constructions, 4 known third-party warnings, 0 unexpected warnings). Posterior rows live in `output/data/pymdp_policy_posterior_grid.json` and must remain normalized (`1`).

Graph-world artifacts are deterministic extension outputs declared in `tracks.yaml` rather than new empirical claims. `simulate_si_graph_world.py` writes summary and trace artifacts for the finite graph path; the regenerated summary reports 4 nodes, 4 steps, and goal-reached flag 1. The topology-trace extension records 4 toy topology traces with agreement flag 1.

Given generative matrices A, B, C, D, pymdp computes state beliefs $q(s)$ via variational inference (`infer_states`). The Agent is configured with planning horizon $H = 2$, which defines the **policy depth** used when constructing candidate policies (logged as `num_policies` in the SI summary artifact; see sec. 11).

The default harness records belief entropy per step; extending to full expected-free-energy policy selection (`infer_policies`) is documented as a follow-on track in sec. 13.

SI artifacts (summary, trace, optional JSONL log) record step count, actions, observations, and belief entropy for sec. 11. Steps recorded: 2. Branching-time and variational formulations of active inference planning are treated here as related planning context [Champion et al., 2021, Nuijten et al., 2026]; the live evidence remains the finite local artifacts `si_policy_grid.json`, `si_efe_terms.json`, and `model_checking_witnesses.json`, not a claim about scalable planning performance.

The `interop` fragment treats the GNN files, JSON views, and ontology bindings as a round-trip contract rather than parallel documentation. `output/data/interop_roundtrip_report.json` records 2 deterministic checks; the manuscript only claims losslessness when `true` is true.

The stricter lint artifacts are adjacent evidence, not new model claims: `output/data/gnn_roundtrip_report.json`, `output/reports/gnn_lint_report.json`, `output/data/ontology_alias_index.json`, and `output/data/ontology_profile_matrix.json` must agree before the interop row passes. A missing GNN variable, duplicate ontology alias, dropped JSON field, shape diff, or dtype diff is therefore a validation failure before rendering.

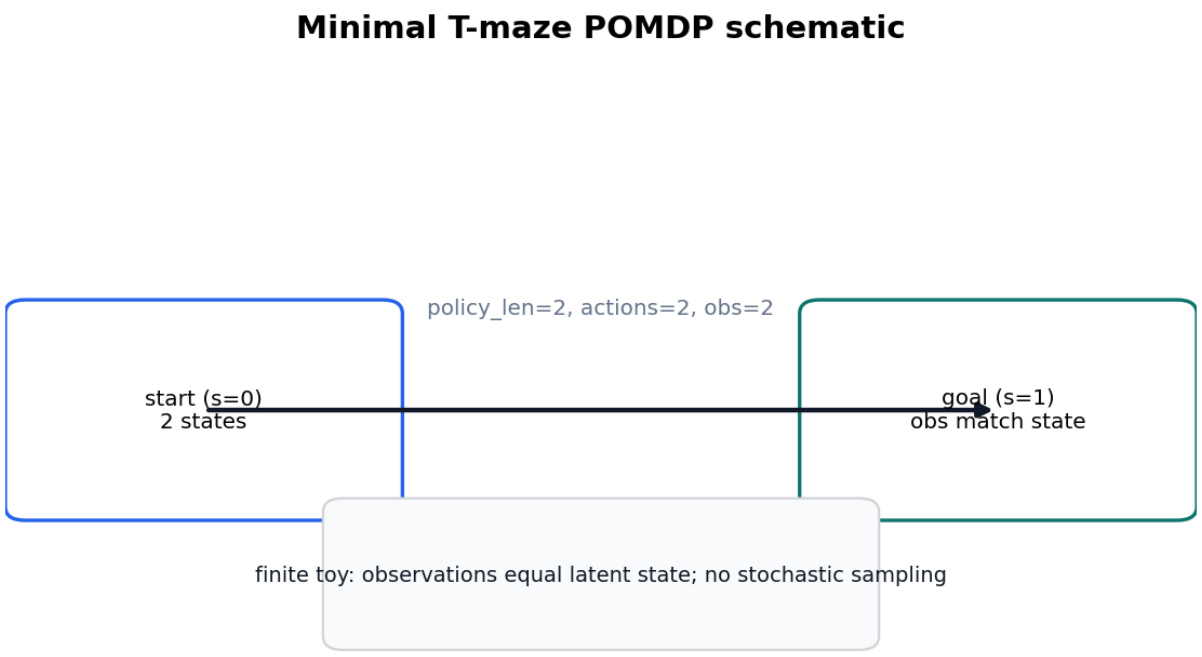


Figure 7: T-maze generative model schematic (2-step policy horizon, state_inference mode).

See `gnn/si_tmaze.gnn.md` for a GNN view of the T-maze hidden state, observation, and policy variables with ontology bindings.

6.0.1 Ontology bindings

- `belief_entropy` → **BeliefEntropy**
- `loc` → **HiddenState**
- `obs` → **ObservationLikelihood**
- `pi` → **PolicyPosterior**

7 Lean formalization boundary

The Lean method is a boundary witness track, not a broad formalization of active inference. `lake build` checks declarations under `lean/TemplateActiveInference/`; fig. 8 renders the proved/deferred surface, while generated inventories carry theorem names, constructive-token status, and axiom checks.

The theorem set links back to the finite analytical and pymdp toys. Horizon witnesses constrain the planning-depth examples, and `efe_additive_identity_from_relations` proves `(risk + ambiguity) + (pragmatic + epistemic) = 0` from the definitional relations using core integer arithmetic (`omega`). These rows join 17 linked theorem-traceability entries with all-linked flag `true`; no prose claim is promoted unless the generated theorem, witness, and evidence-field rows agree.

Lean formalization boundary status

Module	Kind	Name	Status
TemplateActiveInference.Bernoull	def	isingCouplingEntries	proved
TemplateActiveInference.Bernoull	def	isingCouplingSum	proved
TemplateActiveInference.Bernoull	theorem	ising_coupling_sum_zero	proved
TemplateActiveInference.EFEDec	def	efeIdentityResidual	proved
TemplateActiveInference.EFEDec	theorem	efe_additive_identity_from_relatic	proved
TemplateActiveInference.EFEDec	theorem	efe_total_eq_neg_value	proved
TemplateActiveInference.EFEDec	def	efeRiskTerms	proved
TemplateActiveInference.EFEDec	def	efeAmbiguityTerms	proved
TemplateActiveInference.EFEDec	def	efePragmaticTerms	proved
TemplateActiveInference.EFEDec	def	efeEpistemicTerms	proved
TemplateActiveInference.EFEDec	theorem	efe_witness_residual_zero	proved
TemplateActiveInference.Promoti	def	PromotionProof	proved
TemplateActiveInference.Promoti	theorem	promotion_complete_iff_all_fields	proved
TemplateActiveInference.Promoti	theorem	promotion_missing_producer_inc	proved
TemplateActiveInference.Promoti	theorem	promotion_all_fields_complete	proved
TemplateActiveInference.Sophisti	def	defaultPolicyLen	proved
TemplateActiveInference.Sophisti	theorem	sophisticated_requires_horizon	proved
TemplateActiveInference.Sophisti	def	tmazeStep	proved
TemplateActiveInference.Sophisti	theorem	tmaze_two_forward_steps_reach_	proved
TemplateActiveInference.Sophisti	theorem	tmaze_goal_absorbing	proved
TemplateActiveInference.Sophisti	def	graphWorldStep	proved
TemplateActiveInference.Sophisti	theorem	graph_world_three_steps_reach_g	proved
TemplateActiveInference.Sophisti	def	branchGraphWorldStep	proved
TemplateActiveInference.Sophisti	theorem	branch_graph_world_three_steps_	proved
TemplateActiveInference.Sophisti	def	loopGraphWorldStep	proved
TemplateActiveInference.Sophisti	theorem	loop_graph_world_four_steps_rea	proved
TemplateActiveInference.Sophisti	def	diamondGraphWorldStep	proved
TemplateActiveInference.Sophisti	theorem	diamond_graph_world_four_steps	proved
TemplateActiveInference.Sophisti	def	finitePolicies	proved
TemplateActiveInference.Sophisti	theorem	policy_enumeration_contains_for	proved
TemplateActiveInference.Sophisti	def	twoStateBeliefWeights	proved
TemplateActiveInference.Sophisti	theorem	two_state_belief_weights_sum_to	proved
TemplateActiveInference.Sophisti	def	twoPolicyPosteriorWeights	proved
TemplateActiveInference.Sophisti	theorem	two_policy_posterior_weights_sur	proved

Figure 8: Lean formalization boundary: module witnesses checked by lake build.

Lean module `TemplateActiveInference.SophisticatedInference` declares the planning-horizon parameter `defaultPolicyLen` and finite T-maze boundary witnesses: `sophisticated_requires_horizon : defaultPolicyLen > 1`, `tmaze_two_forward_steps_reach_goal`, and `tmaze_goal_absorbing`. It also contains constructive finite witnesses for graph-world reachability, finite policy enumeration, two-state belief weights, and two-policy posterior weights. These theorems formalize small finite boundaries shared with generated artifacts; they do *not* prove that the toy policy posterior is a general model of sophisticated inference. Axioms are audited with `#print axioms` (the gate whitelists only `propext`, `Classical.choice`, `Quot.sound`); see the Lean track gate.

Build via `lake build` under `lean/`.

The `model_checking` fragment complements Lean with finite exhaustive witnesses. `output/reports/model_checking_witnesses.json` records 12 toy-state witnesses and reports `true` only when no counterexample is found in the enumerated state/action space.

This is deliberately narrower than a semantic proof of all Active Inference programs. It checks the finite T-maze and graph-world boundary objects used by this manuscript and exposes the witness inventory to the same artifact and claim gates as the Lean theorem inventory. The Lean graph-world inventory witnesses 4 generated toy topology ids, with all-topologies-witnessed flag `true`; theorem traceability contributes 17 linked rows.

The `theorem_traceability` fragment binds Lean theorem inventory rows to finite model-checking witnesses, manuscript claims, and evidence fields. `output/data/theorem_traceability_matrix.json` records 17 traceability rows and passes only when every theorem row is linked (`true`).

7.0.1 Proof extraction track

The `proof_extraction` track extracts Lean theorem statements and proof-source metadata into `output/data/proof_extraction_index.json`. The index currently contains 12 extracted theorem rows, with constructive-token status `true`.

8 Sheaf composition

8.1 Compose contract

Each manifest row in `manuscript/sheaf/manifest.yaml` binds fragment tracks from `manuscript/sheaf/tracks.yaml`. A track supplies a renderer, compose order, label, optional flag, general paper role, and paper-specific use statement; the composer then flattens the binding set into one Markdown section for PDF and web output.

The operational claim is auditable binding. Analytical, simulation, pymdp, visualization, Lean, GNN, ontology, scholarship, and optional media fragments attach to IMRAD rows under eq. 3 (**P** present, — unbound, **M** missing). This is an applied local-to-global consistency contract in the spirit of cellular sheaf and sheaf-signal-processing work [Curry, 2014, Robinson, 2014], instantiated here as a finite artifact gate rather than a cohomology claim.

8.2 Coverage and figures

fig. 9 summarizes 34 fragment types and their IMRAD bindings. Generated tables below list every track definition and section×track binding at compose time. The visualization track is gated by `output/reports/visualization_quality_audit.json`: 23 / 23 registered figures render, 23 are source-mapped, and 23 have sufficient alt/caption metadata; the all-quality flag is `true`.

The visualization gate is deliberately row-level. It requires declared visual/evidence roles (`true`), artifact-backed paper claims (`true`), section bindings (`true`), RGB nonblank image renders, hashes, and source-map agreement. A registered figure whose render, hash, or source map is absent or mismatched fails the row-level audit (negative control: deleting an output render flips `true` to false during recomposition). Rows missing a declared role still fail the all-quality flag — a known-wrong row is flagged by the audit rather than implicitly satisfied (negative control: `tests/test_figures.py::test_visualization_quality_audit_rejects_stale_section_and_source_rows`). The statistical bridge then expands 7 statistically backed figures into 7 figure-source-scholarship rows with connected status `true`, manuscript-reference status `true`, and visualization-bound reference status `true`. Rows lacking declared roles are rejected outright — an undeclared visualization cannot slip past the gate.

Negative controls for these gates live in `tests/gates/` (e.g. `test_validate_outputs_negative_missing_sheaf_matrix`, `test_figures_nonblank_negative_blank_png`) and `tests/test_track_consolidation_negative.py` asserts forged or missing rows fail closed. The claim ledger is also checked at row level rather than as prose metadata. `claim_evidence_audit.json` resolves 97 claim rows to live artifacts (`true`) and replays their typed predicates (`true`), yielding the promoted completeness flag `true`.

8.3 Compose commands

```
uv run python scripts/compose_manuscript.py
uv run python scripts/compose_manuscript.py --validate-only --strict
```

Each run emits `output/data/sheaf_coverage_matrix.json` and regenerates coverage artifacts. Partial compose (`--section`) is draft-only; the matrix always reflects the full manifest. Coverage totals appear on sec. 1; discussion scope is in sec. 13.

8.4 Law verification

`--validate-only --strict` runs the structural gate before any fragment is glued. Beyond per-cell coverage, it invokes the sheaf-law oracle (`verify_sheaf_laws`, `src/manuscript/sheaf/laws.py`), which checks 6 axioms — poset, presheaf functoriality, separation, gluing, typing, and compositionality — and reports 6/6 satisfied for the current manifest. A violation is raised as an error-level issue and aborts the build, so a malformed manifest (a section colliding on an output file, an off-chain block, a mistyped fragment, a fragment shared between sections) can never compose. The formal statements are in the formalism block below; the negative-control suite (`tests/test_sheaf_laws.py`) proves each check is falsifiable.

The semantic layer is separate from those structural laws. `output/data/sheaf_gluing_certificate.json` records cross-track symbols, typed claim evidence, artifact sources, and manuscript-variable restrictions; validation fails when the analytical, pymdp, GNN, ontology, Lean, visualization, or manuscript tracks disagree about a shared symbol or measured claim. The visualization-quality audit is one of those restrictions, so a missing source map, missing statistical bridge source, missing hash, blank render, non-RGB render, undersized figure, or unbound section breaks the same semantic contract that checks statistics and theorem witnesses. fig. 10 renders the configured producers, generated evidence artifacts, and validation consumers that read each shared symbol.

8.4.1 Base poset and presheaf

The manuscript is modelled as a coverage sheaf over a finite base poset. Let the **base** P be the IMRAD blocks ordered as a chain,

$$\mathsf{Introduction} \prec \mathsf{Methods} \prec \mathsf{Results} \prec \mathsf{Discussion} \prec \mathsf{Appendix}, \tag{2}$$

with, in each block, a *group* node above its *section* nodes (written $G \sqsupseteq s$). P is therefore a finite poset (equivalently a finite Alexandrov space). Let $\mathcal{T}$ be the registered fragment-track set from `manuscript/sheaf/tracks.yaml`; each track $t \in \mathcal{T}$ carries a renderer $R(t)$, label $L(t)$, optional flag $O(t)$, a general paper role $U(t)$, a section-use statement $V(t)$, and a strict compose-order index $\pi(t)$.

The **presheaf** $\mathcal{F}$ is a contravariant functor on P — $\mathcal{F} : P \to \mathbf{Set}$ with restriction maps along $\sqsupseteq$ — assigning to each composing section s its bound fragment set $\mathcal{F}(s) = \{\,(t, F_s(t)) : t \text{ bound in } s\,\}$, where $F_s : \mathcal{T} \rightharpoonup \mathbf{Path}$ is the section's partial binding map. Restriction along $G \sqsupseteq s$ is projection onto a section's own bindings; group nodes carry the empty assignment and do not compose.

The coverage cell is

$$B(s,t) \in \{\mathrm{P}, —, \mathrm{M}\} \tag{3}$$

derived from $F_s(t)$ and filesystem existence at compose time: **P** when a bound fragment exists, — when the track is unbound for that row, and **M** when a bound path is missing. The current regenerated matrix reports 95 present / 95 bound / 0 missing cells. Registry size: $|\mathcal{T}| = 34$ types across 17 IMRAD manifest rows (5 group rows, 12 composing sections).

8.4.2 Verified sheaf laws

What makes this presheaf a *sheaf* — rather than a bare incidence table — is that the composer's structural axioms are machine-checked. The oracle `verify_sheaf_laws` (`src/manuscript/sheaf/laws.py`) verifies 6 laws, and the regenerated build reports 6/6 satisfied:

1. **Poset.** The IMRAD blocks form the chain of eq. 2; compose order is monotone in block rank and every composing section's block carries a group row.
2. **Presheaf (functoriality).** Every bound track lies in $\mathcal{T}$; π is a strict total order; and each section's resolved track order is the monotone restriction of π (an explicit `track_order` override must be a permutation of the section's bound tracks).
3. **Separation (locality).** The map $s \mapsto \mathrm{output_name}(s)$ is injective over composing sections: distinct locals glue to distinct global positions, so the global section is unique.
4. **Gluing.** Compose order is a linear extension of P — each block's rows are contiguous and strictly increasing in order — so the local fragments glue to a unique global manuscript in which every composing section appears exactly once.
5. **Typing.** Each binding $(t, F_s(t))$ is well-typed: $R(t)$ is a registered renderer and the fragment suffix lies in $R(t)$'s accepted suffix set. Generated renderers (`section_figures`, `layers_report`) synthesize their body and are explicitly type-exempt.
6. **Compositionality.** Every fragment file is private to one section (no path is bound twice), so global composition is the coproduct of the per-section bodies and is independent of inclusion order.

Each law is paired with a negative control in `tests/test_sheaf_laws.py` — a single mutation that breaks the law and is proven to be caught — so the gate binds the laws' *content*, not merely their shape. Under `--strict`, any violation is surfaced as an error-level manifest issue and aborts composition.

8.4.3 Scope (what is and is not claimed)

These laws verify the sheaf *axioms* on a finite base poset. They do **not** compute sheaf *cohomology* (H^0/H^1, Čech complexes, derived functors); "sheaf" here names the verified separation-and-gluing structure of a multi-track coverage assignment, not a cohomological invariant. Formal track definitions and section×track bindings appear in the generated tables below.

Semantic gluing then checks agreement of the glued content: coverage counts, manuscript variables, typed claim predicates, pymdp mode/hash, Bernoulli GNN ontology, and SI T-maze GNN ontology. This certificate is a content-level audit over the same base, not an additional topological law.

The `provenance` fragment makes artifact lineage a live canonical sheaf track. The configured producer `generate_sheaf_tracks.py` writes `output/data/artifact_provenance.json`, which hashes 86 required toy artifacts and records producer scripts, source commit, deterministic seed fields, config digests, and 5 artifact bundles. Publication claims that depend on generated files must be traceable to this lineage table or to a narrower artifact-specific certificate.

The provenance claim is intentionally limited: every listed artifact exists, has a SHA-256 digest or an explicit cycle exclusion, is produced by a configured analysis script, and carries seed/config provenance (`86` seeded rows; all seeded flag `true`; bundle-complete flag `true`). A changed file, missing producer, or stale saved digest is a validation failure, not a prose warning.

The `counterexample` fragment records expected-failure fixtures as first-class evidence. `output/reports/counterexample_matrix.json` lists 25 negative controls that intentionally mutate ontology mappings, semantic certificates, graph-world trace agreement, typed claim evidence, replay rows, release parity, and provenance hashes.

The matrix is not an empirical result. It is a falsifiability ledger: each row names the gate that must fail and the test that proves the failure path remains live.

The `adversarial_audit` fragment makes expected failures part of the sheaf rather than an informal test note. `output/reports/adversarial_audit.json` records 25 known-bad rows and 0 known-bad rows passing; publication proceeds only when every row is documented as an expected failure and mapped to a gate.

The audit rows target the same failure modes as the semantic certificate: incomplete sweep cells, unnormalized uncertainty rows, interop field loss, stale certificate state, and empirical-scope leakage. The scope boundary remains toy-only: `toy_only_pass`.

The `evidence_fields` fragment indexes the exact artifact fields that support typed claims and hydrated manuscript tokens. `output/data/evidence_field_index.json` records 97 field rows, and the track passes only when every referenced JSONPath or dotted field is present (`true`).

The `release_bundle` fragment records whether the canonical deliverables exist before copying and whether copied root outputs match or are explicitly deferred until the copy stage. `output/reports/release_bundle_manifest.json` tracks 36 required deliverables with source-present flag `true`.

The bundle contract is now indexed artifact-by-artifact rather than inferred from isolated reports. `output/data/artifact_contract_index.json` contains 86 generated artifact rows; each row binds its producer, configured script, pipeline/sheaf lanes, manuscript consumers, claim predicates, validators, negative control, freshness status, and copied-output parity. The aggregate row-complete flag is `true`, and copied-root parity completeness is `true`.

The `gate_ergonomics` fragment turns validation commands into evidence rows. `output/data/validation_gate_index.json` records 26 gate rows, each naming required inputs and the negative-control surface that should fail closed.

`output/data/track_lane_matrix.json` is the cross-track audit table for the same gate surface: 32 pipeline rows map to sheaf fragments, producer scripts, primary artifacts, validation gates, and manuscript consumers, with completion flag `true`.

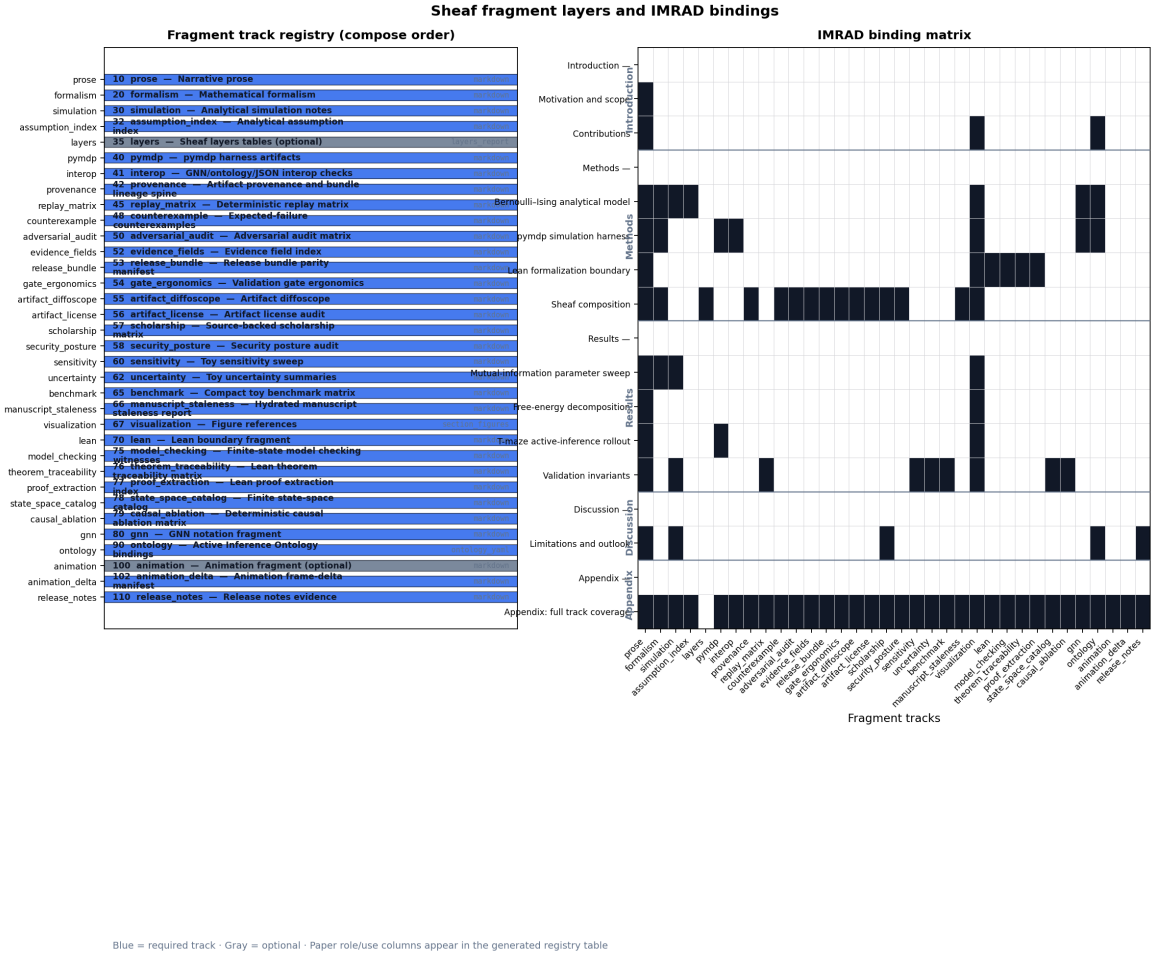


Figure 9: Sheaf layers overview: registry stack (compose order, renderer ids) and IMRAD binding heatmap for 34 fragment types across 17 manifest rows (95 present / 95 bound / 0 missing), generated from `output/data/sheaf_coverage_matrix.json`.

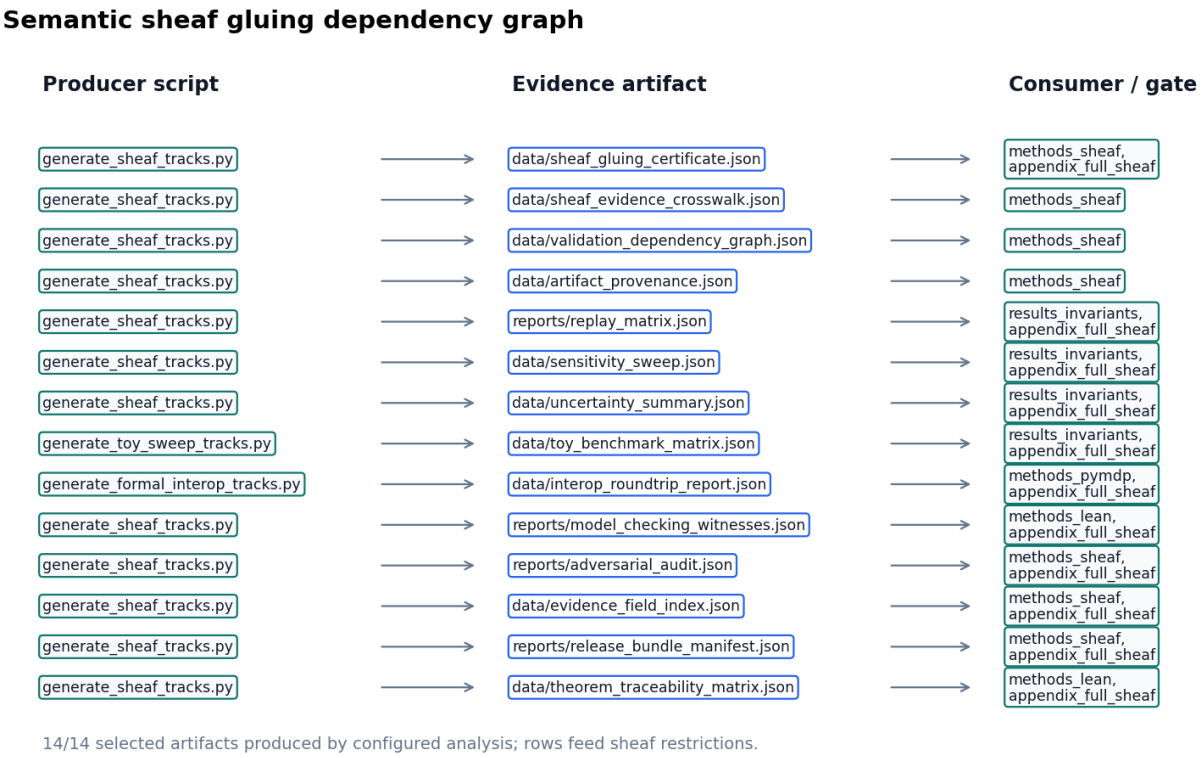


Figure 10: Semantic gluing graph: configured producers, generated evidence artifacts, and validation consumers for the multi-track sheaf certificate.

8.4.4 Artifact diffoscope track

The `artifact_diffoscope` track compares saved provenance hashes against live artifact hashes at the artifact root JSONPath. Its proof artifact is `output/reports/artifact_diffoscope.json`: it currently records 43 comparison rows, with equality status `true`.

8.4.5 Artifact license track

The `artifact_license` track classifies generated and project-source artifacts under the public project license boundary. Its audit artifact is `output/reports/artifact_license_audit.json`: it currently records 86 rows, with license-safe status `true`.

The `scholarship` fragment turns citations into an audited method surface rather than decorative bibliography. `output/data/scholarship_source_matrix.json` records 21 source rows across 21 method roles and 10 source families, including 3 quantitative/statistical or visualization-quality method roles; fig. 13 renders the resulting source-to-artifact map with 1 locator kinds. The row set connects foundational free-energy and active-inference references [Friston, 2010, Buckley et al., 2017, Da Costa et al., 2020, Parr et al., 2022, Smith et al., 2022], planning context [Champion et al., 2021, Nuijten et al., 2026], implementation and notation anchors [Heins et al., 2022, Smékal and Friedman, 2023], and applied sheaf/local-to-global sources [Curry, 2014, Robinson, 2014, Bosca and Ghrist, 2026] to the exact artifact or method role they support.

The validation claim is deliberately narrow: every row must have a bibliography entry with a DOI or URL, a manuscript citation, registered sheaf tracks, bound manifest consumer sections, an existing evidence artifact, and a scope-guarded claim-boundary statement. The saved matrix is then rederived from live bibliography, manuscript, registry, manifest, and artifact evidence before validation accepts it (`true`), so a forged row-level boolean cannot launder a disconnected source. The added statistics and visualization rows point to `analysis_statistics.json` and `visualization_quality_audit.json`, including a statistical-visualization bridge row, so the scholarship track now distinguishes method lineage from the generated numerical, figure-quality, and figure-provenance evidence. The hydrated flags `true`, `true`, and `true` are therefore source-traceability and scope-control claims, not claims that the toy results inherit empirical support from the cited literature.

The newer arXiv rows are intentionally constrained. They situate the toy EFE and planning artifacts against branching-time and variational-planning work, and they situate the finite manuscript sheaf against modern local-to-global computation framing, but none of those citations promotes empirical, neural network, or scalable-agent performance claims for this exemplar.

The security-posture track treats the public exemplar itself as the defended asset. `output/reports/security_posture_audit.json` records 9 controls: 7 enforced local controls and 2 production-security obligations that are explicitly deferred rather than claimed. The enforced rows cover public-data boundaries, offline reproducibility, artifact hashes, copied-output parity, claim/scope traceability, the Lean boundary, and a source/config secret-pattern scan.

The audit is intentionally not a production certification. It records 0 high-risk local gaps and 0 high-risk secret-pattern findings; the all-controls flag is `true`, and all listed evidence is present: `true`. Deferred rows cover signed provenance/SBOM release attestation and zero-trust runtime controls, which require deployment-specific identity, device posture, logging, and signing infrastructure outside

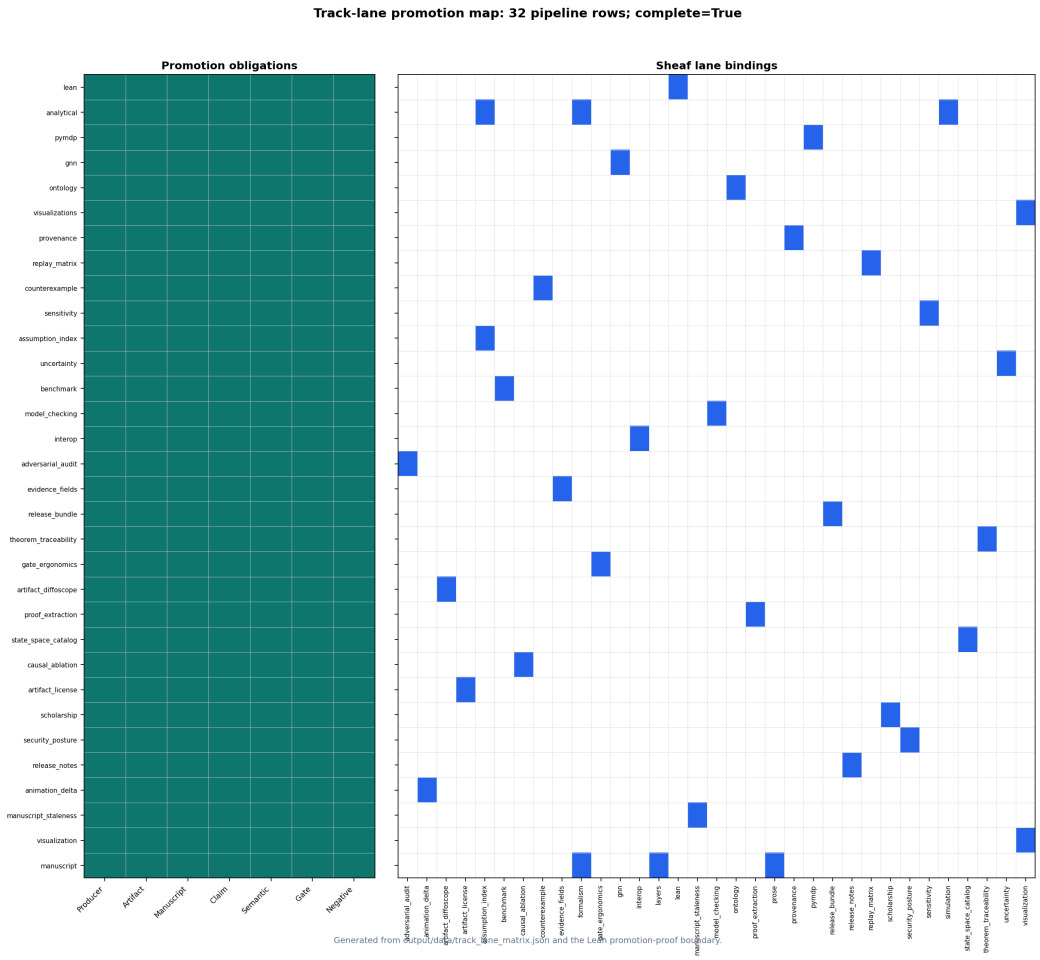


Figure 11: Track-lane promotion map: 32 pipeline-to-sheaf rows with complete promotion status true. Left: seven promotion-rule obligations; right: sheaf fragment bindings.

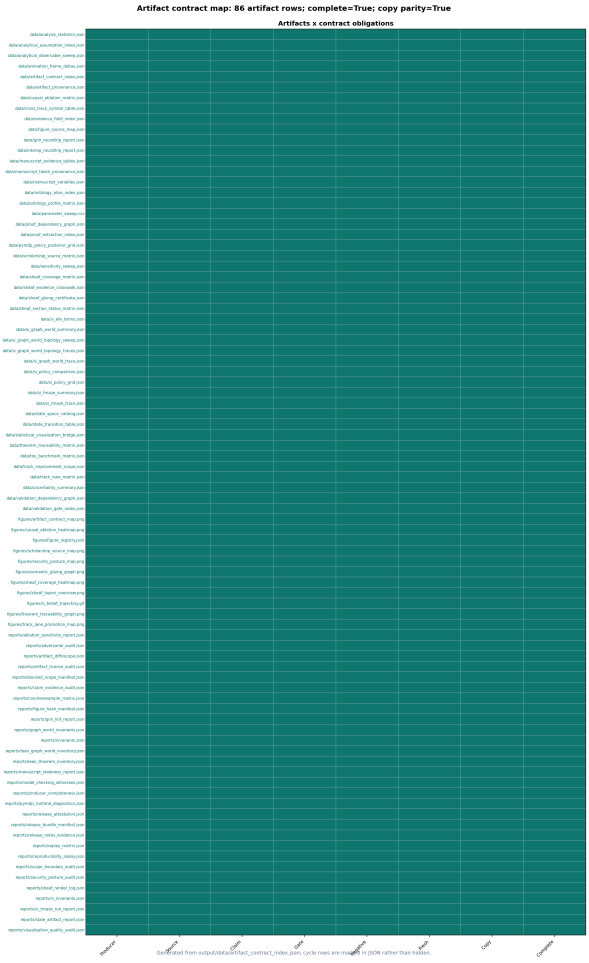


Figure 12: Artifact contract map: 86 generated artifact rows with complete contract status true and copied-output parity complete true. Cycle rows are explicit in `output/data/artifact_contract_index.json`.



Figure 13: Scholarship source map: 21 source rows across 21 method roles and 10 source families. Connected status: true; row evidence rederived: true.

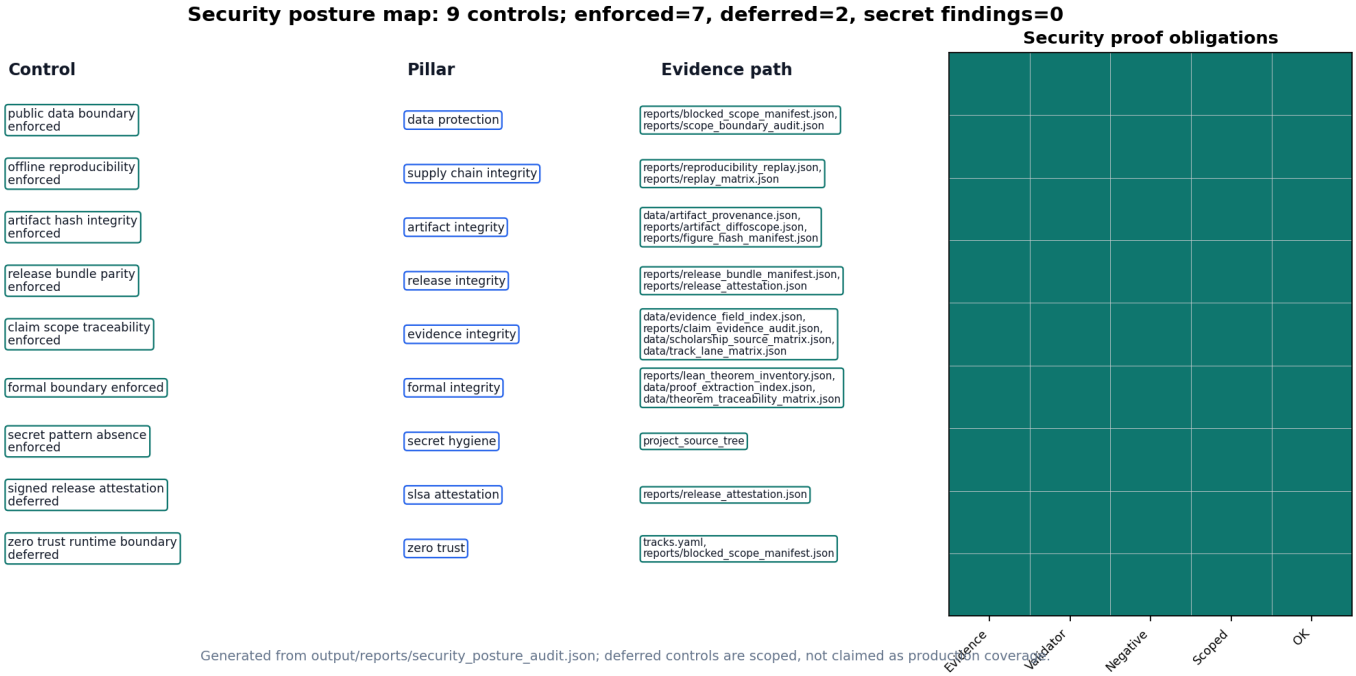


Figure 14: Security posture map: 9 controls, 7 enforced and 2 scoped as deferred; secret findings: 0; high-risk gaps: 0.

this toy-only manuscript.

The `manuscript_staleness` fragment closes the hydration loop. `output/reports/manuscript_staleness_report.json` checks 322 manuscript token bindings against the current generated variables after resolved markdown is written; the pass flag is `true`.

This is a publication-systems claim, not a domain result. A stale hydrated value, unresolved token, or missing resolved section becomes a validation failure before PDF or web outputs are accepted.

8.5 Sheaf fragment track registry

Compose order and renderer bindings from `manuscript/sheaf/tracks.yaml`.

Order	Track id	Label	Renderer	Paper role	Paper use	Optional
10	`prose`	Narrative prose	`markdown`	Narrative framing and argument flow	Supports the narrative spine for each composed paper section.	No
20	`formalism`	Mathematical formalism	`markdown`	Mathematical definitions and equations	States the finite equations, laws, and boundary assumptions used by prose claims.	No
30	`simulation`	Analytical simulation notes	`markdown`	Deterministic toy analysis evidence	Connects analytical sweeps and toy simulations to results claims.	No
32	`assumption_index`	Analytical assumption index	`markdown`	Assumption boundary ledger	Lists finite-model assumptions so analytical claims stay scoped.	No
35	`layers`	Sheaf layers tables	`layers_report`	Registry and binding disclosure	Generates the track registry, binding matrix, and evidence crosswalk tables.	Yes
40	`pymdp`	pymdp harness artifacts	`markdown`	Active-inference implementation evidence	Binds pymdp traces, runtime diagnostics, and policy comparisons to methods and results.	No
41	`interop`	GNN/ontology/JSON interop checks	`markdown`	Cross-format compatibility evidence	Shows that GNN, ontology, and JSON artifacts preserve model meaning.	No
42	`provenance`	Artifact provenance and bundle lineage spine	`markdown`	Artifact lineage evidence	Documents producers, hashes, seeds, and bundle lineage for generated claims.	No

Order	Track id	Label	Renderer	Paper role	Paper use	Optional
45	`replay_matrix`	Deterministic replay matrix	`markdown`	Reproducibility replay evidence	Shows configured producers replay and match their expected artifacts.	No
48	`counterexample`	Expected-failure counterexamples	`markdown`	Falsifiability negative controls	Records known-bad fixtures that must fail validation gates.	No
50	`adversarial_audit`	Adversarial audit matrix	`markdown`	Adversarial robustness evidence	Documents stress cases and expected failures for sheaf-track claims.	No
52	`evidence_fields`	Evidence field index	`markdown`	Claim field traceability	Maps evidence fields to sections and artifacts for claim hydration.	No
53	`release_bundle`	Release bundle parity manifest	`markdown`	Release artifact parity evidence	Checks that required deliverables exist and copied outputs match or defer explicitly.	No
54	`gate_ergonomics`	Validation gate ergonomics	`markdown`	Validation workflow index	Explains the gates a reader or maintainer can rerun locally.	No
55	`artifact_diffoscope`	Artifact diffoscope	`markdown`	Artifact equality evidence	Compares generated and copied artifacts to surface publication drift.	No
56	`artifact_license`	Artifact license audit	`markdown`	License safety evidence	Records license status for artifacts included in release surfaces.	No
57	`scholarship`	Source-backed scholarship matrix	`markdown`	Scholarship and method-source lineage	Connects cited sources to method roles, sections, and generated evidence.	No
58	`security_posture`	Security posture audit	`markdown`	Public release security boundary evidence	Separates enforced local controls from deferred production-security obligations.	No

Order	Track id	Label	Renderer	Paper role	Paper use	Optional
60	sensitivity	Toy sensitivity sweep	markdown	Parameter sensitivity evidence	Summarizes deterministic toy perturbations behind robustness claims.	No
62	uncertainty	Toy uncertainty summaries	markdown	Uncertainty summary evidence	Reports normalized uncertainty bins and summaries for finite toy analyses.	No
65	benchmark	Compact toy benchmark matrix	markdown	Toy benchmark comparison evidence	Shows compact model comparisons used to bound toy-only claims.	No
66	manuscript_staleness	Hydrated manuscript staleness report	markdown	Manuscript freshness evidence	Checks hydrated sections against current generated artifacts and variables.	No
67	visualization	Figure references	section_figures	Figure evidence and communication	Injects registry figures into section-specific evidence blocks.	No
70	lean	Lean boundary fragment	markdown	Formal proof boundary evidence	Separates proved Lean witnesses from intentionally scoped formal boundaries.	No
75	model_checking	Finite-state model checking witnesses	markdown	Exhaustive finite-model evidence	Lists model-checking witnesses for finite state-space claims.	No
76	theorem_traceability	Lean theorem traceability matrix	markdown	Theorem dependency traceability	Links theorem rows to proof dependencies and finite model witnesses.	No
77	proof_extraction	Lean proof extraction index	markdown	Constructive proof extraction evidence	Shows extracted theorem artifacts remain constructive and available.	No
78	state_space_catalog	Finite state-space catalog	markdown	Finite model catalog evidence	Enumerates reachable states so toy models remain explicitly finite.	No

Order	Track id	Label	Renderer	Paper role	Paper use	Optional
79	causal_ablation	Deterministic causal ablation matrix	markdown	Causal ablation evidence	Summarizes deterministic perturbation effects across toy topologies.	No
80	gnn	GNN notation fragment	markdown	GNN notation evidence	Documents notation and round-trip status for the analytical model.	No
90	ontology	Active Inference Ontology bindings	ontology_yaml	Ontology binding evidence	Maps local variables to ontology terms for semantic consistency.	No
100	animation	Animation fragment	markdown	Dynamic trace visualization	Provides a deterministic GIF trace as optional appendix evidence.	Yes
102	animation_delta	Animation frame-delta manifest	markdown	Animation integrity evidence	Confirms animation frames change and support the visual trace.	No
110	release_notes	Release notes evidence	markdown	Release narrative evidence	Binds release-note statements to source-backed artifacts.	No

Track count: 34 registered fragment types.

8.6 IMRAD binding matrix

Compact section summary: **P** = present (bound and file exists); **M** = missing (bound, file absent). Every track not listed for a row is absent (not bound).

The complete cell-level matrix is stored in `output/data/sheaf_coverage_matrix.json` and visualized in fig. 3.

Section	Present tracks (P)	Missing tracks (M)
Introduction (group)	—	—
Motivation and scope	prose	—
Contributions	prose, visualization, ontology	—
Methods (group)	—	—
Bernoulli–Ising analytical model	prose, formalism, simulation, assumption_index, visualization, gnn, ontology	—
pymdp simulation harness	prose, formalism, pymdp, interop, visualization, gnn, ontology	—
Lean formalization boundary	prose, visualization, lean, model_checking, theorem_traceability, proof_extraction	—
Sheaf composition	prose, formalism, layers, provenance, counterexample, adversarial_audit, evidence_fields, release_bundle, gate_ergonomics, artifact_diffoscope, artifact_license, scholarship, security_posture, manuscript_staleness, visualization	—
Results (group)	—	—

Section	Present tracks (P)	Missing tracks (M)
Mutual-information parameter sweep	prose, formalism, simulation, visualization	—
Free-energy decomposition	prose, visualization	—
T-maze active-inference rollout	prose, pymdp, visualization	—
Validation invariants	prose, simulation, replay_matrix, sensitivity, uncertainty, benchmark, visualization, state_space_catalog, causal_ablation	—
Discussion (group)	—	—
Limitations and outlook	prose, simulation, scholarship, ontology, release_notes	—
Appendix (group)	—	—
Appendix: full track coverage	prose, formalism, simulation, assumption_index, pymdp, interop, provenance, replay_matrix, counterexample, adversarial_audit, evidence_fields, release_bundle, gate_ergonomics, artifact_diffoscope, artifact_license, scholarship, security_posture, sensitivity, uncertainty, benchmark, manuscript_staleness, visualization, lean, model_checking, theorem_traceability, proof_extraction, state_space_catalog, causal_ablation, gnn, ontology, animation, animation_delta, release_notes	—

Totals: 95 present / 95 bound / 0 missing.

Symbol	Coverage color	Meaning
P	Black	Track **present** (bound and fragment exists)
—	White	**Absent** (not bound for this section)
M	Gray	**Missing** (bound but fragment file absent)

8.7 Section-track status

Generated status for the current manuscript sheaf, summarized per composable section.

Section	IMRAD	Bound	Present	Missing	Status
Motivation and scope	introduction	1	1	0	fully_sheafed
Contributions	introduction	3	3	0	fully_sheafed
Bernoulli–Ising analytical model	methods	7	7	0	fully_sheafed
pymdp simulation harness	methods	7	7	0	fully_sheafed
Lean formalization boundary	methods	6	6	0	fully_sheafed
Sheaf composition	methods	15	15	0	fully_sheafed
Mutual-information parameter sweep	results	4	4	0	fully_sheafed
Free-energy decomposition	results	2	2	0	fully_sheafed
T-maze active-inference rollout	results	3	3	0	fully_sheafed

Section	IMRAD	Bound	Present	Missing	Status
Validation invariants	results	9	9	0	fully_sheafed
Limitations and outlook	discussion	5	5	0	fully_sheafed
Appendix: full track coverage	appendix	33	33	0	fully_sheafed

Section status: 12 / 12 composable sections fully sheafed; 0 required bound fragments missing.

8.8 Track status

Track	Renderer	Bound sections	Present	Missing	Claims	Status
prose	markdown	12	12	0	0	complete
formalism	markdown	5	5	0	0	complete
simulation	markdown	5	5	0	11	complete
assumption_index	markdown	2	2	0	1	complete
layers	layers_report	1	1	0	1	complete
pymdp	markdown	3	3	0	15	complete
interop	markdown	2	2	0	3	complete
provenance	markdown	2	2	0	15	complete
replay_matrix	markdown	2	2	0	3	complete
counterexample	markdown	2	2	0	2	complete
adversarial_audit	markdown	2	2	0	9	complete
evidence_fields	markdown	2	2	0	1	complete
release_bundle	markdown	2	2	0	9	complete
gate_ergonomics	markdown	2	2	0	8	complete
artifact_diffoscope	markdown	2	2	0	2	complete
artifact_license	markdown	2	2	0	1	complete
scholarship	markdown	3	3	0	4	complete
security_posture	markdown	2	2	0	2	complete
sensitivity	markdown	2	2	0	9	complete
uncertainty	markdown	2	2	0	4	complete
benchmark	markdown	2	2	0	3	complete
manuscript_staleness	markdown	2	2	0	1	complete
visualization	section_figures	10	10	0	17	complete
lean	markdown	2	2	0	9	complete
model_checking	markdown	2	2	0	7	complete
theorem_traceability	markdown	2	2	0	3	complete
proof_extraction	markdown	2	2	0	2	complete
state_space_catalog	markdown	2	2	0	2	complete
causal_ablation	markdown	2	2	0	2	complete
gnn	markdown	3	3	0	4	complete
ontology	ontology_yaml	5	5	0	5	complete
animation	markdown	1	1	0	2	complete
animation_delta	markdown	1	1	0	1	complete
release_notes	markdown	2	2	0	2	complete

Status cells: 578 section-track cells.

8.9 Render and logging summary

Event	Component	Output	Status	Detail
registry_loaded	sheaf.registry	registered_tracks	ok	34 tracks
manifest_loaded	sheaf.manifest	manifest_sections	ok	17 sections
coverage_matrix_built	sheaf.coverage	output/data/sheaf_coverage_matrix.json	ok	95 present cells
section_status_matrix_built	sheaf.status	output/data/sheaf_section_status_matrix.json	ok	578 section-track cells
layers_renderer_bound	sheaf.layers_report	manuscript/08_methods_sheaf.md	ok	methods sheaf layer tables
semantic_artifacts_indexed	sheaf.semantic	output/data/validation_dependency_graph.json	ok	86 artifact producer rows
validation_gates_indexed	gates	output/data/validation_gate_index.json	ok	3 gate groups
manuscript_sections_composed	sheaf.compose	manuscript/*.md	ok	16 composed markdown files

Render events: 8.

8.10 Evidence crosswalk

Claim	Artifact	Producer	Gates
sheaf_registry	manuscript/sheaf/tracks.yaml	manual	validate_outputs
sheaf_manifest	manuscript/sheaf/manifest.yaml	manual	validate_outputs
sheaf_coverage_config	manuscript/sheaf/coverage.yaml	manual	validate_outputs
sheaf_coverage_matrix	output/data/sheaf_coverage_matrix.json	generate_figures.py	validate_outputs, validate_manuscript
sheaf_gluing_certificate	output/data/sheaf_gluing_certificate.json	generate_sheaf_tracks.py	validate_manuscript, validate_outputs
sheaf_evidence_crosswalk	output/data/sheaf_evidence_crosswalk.json	generate_sheaf_tracks.py	validate_manuscript, validate_outputs
validation_dependency_graph	output/data/validation_dependency_graph.json	generate_sheaf_tracks.py	validate_manuscript, validate_outputs
semantic_gluing_graph_figure	../figures/semantic_gluing_graph.png	generate_figures.py	validate_outputs, figure_registry

Claim rows: 97 typed evidence claims.

8.11 Artifact producer graph

Artifact	Producer	Configured	Consumers
output/data/analysis_statistics.json	compute_statistics.py	Yes	results_si_tmaze, results_invariants
output/data/analytical_assumption_index.json	generate_toy_sweep_tracks.py	Yes	methods_analytical, appendix_full_sheaf
output/data/analytical_observable_sweep.json	generate_toy_sweep_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/animation_frame_deltas.json	render_animation.py	Yes	appendix_full_sheaf
output/data/artifact_contract_index.json	generate_sheaf_tracks.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/artifact_provenance.json	generate_sheaf_tracks.py	Yes	methods_sheaf
output/data/causal_ablation_matrix.json	generate_toy_sweep_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/cross_track_symbol_table.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf

Artifact	Producer	Configured	Consumers
output/data/evidence_field_index.json	generate_sheaf_tracks.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/figure_source_map.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/gnn_roundtrip_report.json	generate_formal_interop_tracks.py	Yes	methods_pymdp, appendix_full_sheaf
output/data/interop_roundtrip_report.json	generate_formal_interop_tracks.py	Yes	methods_pymdp, appendix_full_sheaf
output/data/manuscript_evidence_tables.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/manuscript_token_provenance.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/manuscript_variables.json	z_generate_manuscript_variables.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/ontology_alias_index.json	generate_formal_interop_tracks.py	Yes	methods_pymdp, appendix_full_sheaf
output/data/ontology_profile_matrix.json	generate_formal_interop_tracks.py	Yes	methods_pymdp, appendix_full_sheaf
output/data/parameter_sweep.csv	run_analytical_sweep.py	Yes	methods_analytical, results_mi_sweep
output/data/proof_dependency_graph.json	generate_sheaf_tracks.py	Yes	methods_lean, appendix_full_sheaf
output/data/proof_extraction_index.json	generate_formal_interop_tracks.py	Yes	methods_lean, appendix_full_sheaf
output/data/pymdp_policy_posterior_grid.json	simulate_si_tmaze.py	Yes	methods_pymdp, appendix_full_sheaf
output/data/scholarship_source_matrix.json	generate_sheaf_tracks.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/sensitivity_sweep.json	generate_sheaf_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/sheaf_coverage_matrix.json	generate_figures.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/sheaf_evidence_crosswalk.json	generate_sheaf_tracks.py	Yes	methods_sheaf
output/data/sheaf_gluing_certificate.json	generate_sheaf_tracks.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/sheaf_section_status_matrix.json	generate_sheaf_tracks.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/si_efe_terms.json	generate_toy_sweep_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/si_graph_world_summary.json	simulate_si_graph_world.py	Yes	methods_pymdp, results_si_tmaze
output/data/si_graph_world_topology_sweep.json	generate_toy_sweep_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/si_graph_world_topology_traces.json	generate_toy_sweep_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/si_graph_world_trace.json	simulate_si_graph_world.py	Yes	methods_pymdp, results_si_tmaze, appendix_full_sheaf
output/data/si_policy_comparison.json	simulate_si_tmaze.py	Yes	methods_pymdp, results_si_tmaze
output/data/si_policy_grid.json	generate_toy_sweep_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/si_tmaze_summary.json	simulate_si_tmaze.py	Yes	methods_pymdp, results_si_tmaze
output/data/si_tmaze_trace.json	simulate_si_tmaze.py	Yes	methods_pymdp, results_si_tmaze
output/data/state_space_catalog.json	generate_toy_sweep_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/state_transition_table.json	generate_sheaf_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/statistical_visualization_bridge.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf

Artifact	Producer	Configured	Consumers
output/data/theorem_traceability_matrix.json	generate_sheaf_tracks.py	Yes	methods_lean, appendix_full_sheaf
output/data/toy_benchmark_matrix.json	generate_toy_sweep_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/track_improvement_scope.json	generate_sheaf_tracks.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/track_lane_matrix.json	generate_sheaf_tracks.py	Yes	methods_sheaf, appendix_full_sheaf
output/data/uncertainty_summary.json	generate_sheaf_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/data/validation_dependency_graph.json	generate_sheaf_tracks.py	Yes	methods_sheaf
output/data/validation_gate_index.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
../figures/si_belief_trajectory.gif	render_animation.py	Yes	appendix_full_sheaf
output/reports/ablation_sensitivity_report.json	generate_sheaf_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/reports/adversarial_audit.json	generate_sheaf_tracks.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/artifact_diffoscope.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/artifact_license_audit.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/blocked_scope_manifest.json	generate_sheaf_tracks.py	Yes	methods_sheaf, discussion_outlook, appendix_full_sheaf
output/reports/claim_evidence_audit.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/counterexample_matrix.json	generate_sheaf_tracks.py	Yes	methods_sheaf
output/reports/figure_hash_manifest.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/gnn_lint_report.json	generate_formal_interop_tracks.py	Yes	methods_pymdp, appendix_full_sheaf
output/reports/graph_world_invariants.json	generate_toy_sweep_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/reports/invariants.json	run_analytical_sweep.py	Yes	results_invariants
output/reports/lean_graph_world_inventory.json	generate_formal_interop_tracks.py	Yes	methods_lean, appendix_full_sheaf
output/reports/lean_theorem_inventory.json	generate_formal_interop_tracks.py	Yes	methods_lean, appendix_full_sheaf
output/reports/manuscript_staleness_report.json	z_generate_manuscript_variables.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/model_checking_witnesses.json	generate_sheaf_tracks.py	Yes	methods_lean, appendix_full_sheaf
output/reports/producer_completeness.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/pymdp_runtime_diagnostics.json	simulate_si_tmaze.py	Yes	methods_pymdp, appendix_full_sheaf
output/reports/release_attestation.json	generate_sheaf_tracks.py	Yes	discussion_outlook, appendix_full_sheaf
output/reports/release_bundle_manifest.json	generate_sheaf_tracks.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/release_notes_evidence.json	generate_integration_audit.py	Yes	discussion_outlook, appendix_full_sheaf
output/reports/replay_matrix.json	generate_sheaf_tracks.py	Yes	results_invariants, appendix_full_sheaf
output/reports/reproducibility_replay.json	generate_validation_spine.py	Yes	results_invariants
output/reports/scope_boundary_audit.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf

Artifact	Producer	Configured	Consumers
output/reports/security_posture_audit.json	generate_sheaf_tracks.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/sheaf_render_log.json	generate_sheaf_tracks.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/si_invariants.json	simulate_si_tmaze.py	Yes	results_si_tmaze
output/reports/si_tmaze_run_report.json	simulate_si_tmaze.py	Yes	results_si_tmaze
output/reports/stale_artifact_report.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf
output/reports/visualization_quality_audit.json	generate_integration_audit.py	Yes	methods_sheaf, appendix_full_sheaf

Producer issues: 0.

8.12 Semantic gluing restrictions

Restriction	Value
Coverage missing	0
Policy comparison rows	4
Policy grid complete	True
Policy posterior rows	10
Policy posterior normalized	True
Runtime unexpected warnings	0
Graph-world trace agrees	True
Animation frames	4
Lean all proved	True
GNN ontology ok	True
Configured producers ok	True
Semantic certificate ok	None
Dependency edges ok	True
Track scope complete	True
Empirical adapter blocked	True
Provenance bundles complete	True
Replay rows matched	True
Sensitivity complete	True
Uncertainty normalized	True
Evidence fields mapped	True
Release bundle sources present	True
Theorem traceability linked	True
Gate ergonomics indexed	True
Interop lossless	True
Scope toy-only	True

8.13 Track-lane matrix

Every pipeline row is shown below in canonical order. The source-owned full matrix in output/data/track_lane_matrix.json remains the authority for exact source paths, claim IDs, semantic-restriction IDs, gate IDs, negative-control IDs, and promotion cells.

Pipeline track	Source, producer, and primary artifact	Coverage, gates, and negative control
lean	Sources (present): TemplateActiveInference. Producer (configured): generate formal interop tracks.py. Primary artifact (present): lean theorem inventory.json.	Role: Lean boundary fragment. Required; matrix complete. Fragments (registered): lean; 2 manuscript consumers; 9 claims; 2 semantic restrictions. Gates: build lean, validate outputs. Negative control: track lane matrix row only forgery.

Pipeline track	Source, producer, and primary artifact	Coverage, gates, and negative control
analytical	Sources (present): analytical. Producer (configured): run analytical sweep.py. Primary artifact (present): parameter sweep.csv.	Role: Python analytical companion. Required; matrix complete. Fragments (registered): formalism, simulation, assumption index; 7 manuscript consumers; 3 claims; 2 semantic restrictions. Gates: validate outputs. Negative control: track lane matrix row only forgery.
pymdp	Sources (present): simulation. Producer (configured): simulate si tmaze.py. Primary artifact (present): si policy comparison.json.	Role: pymdp sophisticated-inference harness. Required; matrix complete. Fragments (registered): pymdp; 3 manuscript consumers; 15 claims; 2 semantic restrictions. Gates: validate outputs. Negative control: track lane matrix row only forgery.
gnn	Sources (present): gnn, gnn. Producer (configured): generate formal interop tracks.py. Primary artifact (present): gnn lint report.json.	Role: Generalized Notation Notation. Required; matrix complete. Fragments (registered): gnn; 3 manuscript consumers; 4 claims; 2 semantic restrictions. Gates: validate outputs. Negative control: track lane matrix row only forgery.
ontology	Sources (present): ontology. Producer (configured): generate formal interop tracks.py. Primary artifact (present): ontology profile matrix.json.	Role: Active Inference Ontology concordance. Required; matrix complete. Fragments (registered): ontology; 5 manuscript consumers; 5 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: track lane matrix row only forgery.
visualizations	Sources (present): visualizations. Producer (configured): generate integration audit.py. Primary artifact (present): visualization quality audit.json.	Role: Figures and optional animation. Required; matrix complete. Fragments (registered): visualization; 10 manuscript consumers; 2 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: track lane matrix row only forgery.
provenance	Sources (present): roadmap tracks, artifact provenance.json. Producer (configured): generate sheaf tracks.py. Primary artifact (present): artifact provenance.json.	Role: Artifact provenance and bundle-lineage spine. Required; matrix complete. Fragments (registered): provenance; 2 manuscript consumers; 15 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: missing sheaf track producer.
replay matrix	Sources (present): roadmap tracks, replay matrix.json. Producer (configured): generate sheaf tracks.py. Primary artifact (present): replay matrix.json.	Role: Deterministic replay matrix. Required; matrix complete. Fragments (registered): replay matrix; 2 manuscript consumers; 3 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: replay mismatch.
counterexample	Sources (present): roadmap tracks, counterexample matrix.json. Producer (configured): generate sheaf tracks.py. Primary artifact (present): counterexample matrix.json.	Role: Expected-failure counterexample matrix. Required; matrix complete. Fragments (registered): counterexample; 2 manuscript consumers; 2 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: known bad counterexample passed.

Pipeline track	Source, producer, and primary artifact	Coverage, gates, and negative control
sensitivity	Sources (present): roadmap tracks, sensitivity sweep.json. Producer (configured): generate sheaf tracks.py. Primary artifact (present): sensitivity sweep.json.	Role: Deterministic toy sensitivity sweep. Required; matrix complete. Fragments (registered): sensitivity; 2 manuscript consumers; 9 claims; 2 semantic restrictions. Gates: validate outputs. Negative control: missing sensitivity cell.
assumption index	Sources (present): roadmap tracks, analytical assumption index.json. Producer (configured): generate toy sweep tracks.py. Primary artifact (present): analytical assumption index.json.	Role: Analytical equation assumption index. Required; matrix complete. Fragments (registered): assumption index; 2 manuscript consumers; 1 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: track lane matrix row only forgery.
uncertainty	Sources (present): roadmap tracks, uncertainty summary.json. Producer (configured): generate sheaf tracks.py. Primary artifact (present): uncertainty summary.json.	Role: Posterior and policy uncertainty summaries. Required; matrix complete. Fragments (registered): uncertainty; 2 manuscript consumers; 4 claims; 2 semantic restrictions. Gates: validate outputs. Negative control: unnormalized uncertainty row.
benchmark	Sources (present): roadmap tracks, toy benchmark matrix.json. Producer (configured): generate toy sweep tracks.py. Primary artifact (present): toy benchmark matrix.json.	Role: Compact deterministic toy benchmark matrix. Required; matrix complete. Fragments (registered): benchmark; 2 manuscript consumers; 3 claims; 2 semantic restrictions. Gates: validate outputs. Negative control: track lane matrix row only forgery.
model checking	Sources (present): roadmap tracks, model checking witnesses.json. Producer (configured): generate sheaf tracks.py. Primary artifact (present): model checking witnesses.json.	Role: Finite-state model-checking witnesses. Required; matrix complete. Fragments (registered): model checking; 2 manuscript consumers; 7 claims; 2 semantic restrictions. Gates: validate outputs. Negative control: missed model checking counterexample.
interop	Sources (present): roadmap tracks, interop roundtrip report.json. Producer (configured): generate formal interop tracks.py. Primary artifact (present): interop roundtrip report.json.	Role: GNN/ontology/JSON round-trip checks. Required; matrix complete. Fragments (registered): interop; 2 manuscript consumers; 3 claims; 2 semantic restrictions. Gates: validate outputs. Negative control: interop shape loss.
adversarial audit	Sources (present): roadmap tracks, adversarial audit.json. Producer (configured): generate sheaf tracks.py. Primary artifact (present): adversarial audit.json.	Role: Expected-failure adversarial audit matrix. Required; matrix complete. Fragments (registered): adversarial audit; 2 manuscript consumers; 9 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: adversarial known bad passes.
evidence fields	Sources (present): roadmap tracks, evidence field index.json. Producer (configured): generate sheaf tracks.py. Primary artifact (present): evidence field index.json.	Role: Artifact field to claim and token index. Required; matrix complete. Fragments (registered): evidence fields; 2 manuscript consumers; 1 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: missing typed claim.
release bundle	Sources (present): roadmap tracks, release bundle manifest.json. Producer (configured): generate sheaf tracks.py. Primary artifact (present): release bundle manifest.json.	Role: Release bundle parity manifest. Required; matrix complete. Fragments (registered): release bundle; 2 manuscript consumers; 9 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: release bundle parity failure.

Pipeline track	Source, producer, and primary artifact	Coverage, gates, and negative control
theorem traceability	Sources (present): roadmap tracks, theorem traceability matrix.json. Producer (configured): generate sheaf tracks.py. Primary artifact (present): theorem traceability matrix.json.	Role: Lean theorem and model-checking traceability matrix. Required; matrix complete. Fragments (registered): theorem traceability; 2 manuscript consumers; 3 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: theorem traceability unlinked.
gate ergonomics	Sources (present): roadmap tracks, validation gate index.json. Producer (configured): generate integration audit.py. Primary artifact (present): validation gate index.json.	Role: Validation gate command/input/negative-control index. Required; matrix complete. Fragments (registered): gate ergonomics; 2 manuscript consumers; 8 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: gate ergonomics unindexed.
artifact diffoscope	Sources (present): roadmap tracks, artifact diffoscope.json. Producer (configured): generate integration audit.py. Primary artifact (present): artifact diffoscope.json.	Role: Artifact hash and JSONPath diffoscope. Required; matrix complete. Fragments (registered): artifact diffoscope; 2 manuscript consumers; 2 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: artifact diffoscope missed hash drift.
proof extraction	Sources (present): roadmap tracks, proof extraction index.json. Producer (configured): generate formal interop tracks.py. Primary artifact (present): proof extraction index.json.	Role: Lean proof extraction evidence index. Required; matrix complete. Fragments (registered): proof extraction; 2 manuscript consumers; 2 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: proof extraction missing statement.
state space catalog	Sources (present): roadmap tracks, state space catalog.json. Producer (configured): generate toy sweep tracks.py. Primary artifact (present): state space catalog.json.	Role: Finite toy state-space catalog. Required; matrix complete. Fragments (registered): state space catalog; 2 manuscript consumers; 2 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: state space catalog missing finite space.
causal ablation	Sources (present): roadmap tracks, causal ablation matrix.json. Producer (configured): generate toy sweep tracks.py. Primary artifact (present): causal ablation matrix.json.	Role: Deterministic toy causal ablation matrix. Required; matrix complete. Fragments (registered): causal ablation; 2 manuscript consumers; 2 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: causal ablation missing cell.
artifact license	Sources (present): roadmap tracks, artifact license audit.json. Producer (configured): generate integration audit.py. Primary artifact (present): artifact license audit.json.	Role: Generated artifact license audit. Required; matrix complete. Fragments (registered): artifact license; 2 manuscript consumers; 1 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: artifact license unsafe artifact.
scholarship	Sources (present): roadmap tracks, scholarship source matrix.json. Producer (configured): generate sheaf tracks.py. Primary artifact (present): scholarship source matrix.json.	Role: Source-backed scholarship matrix. Required; matrix complete. Fragments (registered): scholarship; 3 manuscript consumers; 4 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: missing scholarship source binding.

Pipeline track	Source, producer, and primary artifact	Coverage, gates, and negative control
security posture	Sources (present): roadmap tracks, security posture audit.json. Producer (configured): generate sheaf tracks.py. Primary artifact (present): security posture audit.json.	Role: Local security posture and deferred production controls. Required; matrix complete. Fragments (registered): security posture; 2 manuscript consumers; 2 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: security posture aggregate forgery.
release notes	Sources (present): roadmap tracks, release notes evidence.json. Producer (configured): generate integration audit.py. Primary artifact (present): release notes evidence.json.	Role: Source-backed release notes evidence. Required; matrix complete. Fragments (registered): release notes; 2 manuscript consumers; 2 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: release notes claim failed gate passed.
animation delta	Sources (present): visualizations, animation frame deltas.json. Producer (configured): render animation.py. Primary artifact (present): animation frame deltas.json.	Role: Animation frame-delta manifest. Required; matrix complete. Fragments (registered): animation delta; 1 manuscript consumers; 1 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: track lane matrix row only forgery.
manuscript staleness	Sources (present): roadmap tracks, manuscript staleness report.json. Producer (configured): z generate manuscript variables.py. Primary artifact (present): manuscript staleness report.json.	Role: Hydrated manuscript staleness audit. Required; matrix complete. Fragments (registered): manuscript staleness; 2 manuscript consumers; 1 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: track lane matrix row only forgery.
visualization	Sources (present): visualizations. Producer (configured): generate integration audit.py. Primary artifact (present): visualization quality audit.json.	Role: Sheaf fragment alias for figures. Optional; matrix complete. Fragments (registered): visualization; 10 manuscript consumers; 17 claims; 2 semantic restrictions. Gates: validate manuscript, validate outputs. Negative control: track lane matrix row only forgery.
manuscript	Sources (present): manuscript, sheaf, figures.yaml. Producer (configured): compose manuscript.py. Primary artifact (present): manifest.yaml.	Role: Sheaf composer and registry. Required; matrix complete. Fragments (registered): prose, formalism, layers; 12 manuscript consumers; 44 claims; 2 semantic restrictions. Gates: validate manuscript. Negative control: track lane matrix row only forgery.

Pipeline rows: 32 total; 31 required; every row shown.

8.14 Track improvement scope

Every roadmap row is shown below in canonical order. The source-owned full matrix in output/data/track_improvement_scope.json remains the authority for exact priorities, scope text, gate or predicate IDs, negative-control IDs, and promotion fields.

Track	Evidence progression	Status, gate, and negative control
adversarial audit	Producer: generate sheaf tracks.py. Current and next evidence: adversarial audit.json.	Status: live; priority: high. Gate or predicate: validate outputs, validate manuscript. Negative control: adversarial known bad passes. Scope: deterministic toy-only.

Track	Evidence progression	Status, gate, and negative control
animation	Producer: render animation.py. Current and next evidence: si belief trajectory.gif.	Status: optional; priority: medium. Gate or predicate: validate outputs. Negative control: missing fragment coverage. Scope: deterministic toy-only.
animation delta	Producer: render animation.py. Current and next evidence: animation frame deltas.json.	Status: live; priority: medium. Gate or predicate: validate outputs, validate manuscript. Negative control: missing fragment coverage. Scope: deterministic toy-only.
artifact diffoscope	Producer: generate integration audit.py. Current and next evidence: artifact diffoscope.json.	Status: live; priority: high. Gate or predicate: validate outputs, validate manuscript. Negative control: artifact diffoscope missed hash drift. Scope: deterministic toy-only.
artifact license	Producer: generate integration audit.py. Current and next evidence: artifact license audit.json.	Status: live; priority: high. Gate or predicate: validate outputs, validate manuscript. Negative control: artifact license unsafe artifact. Scope: deterministic toy-only.
assumption index	Producer: generate toy sweep tracks.py. Current and next evidence: analytical assumption index.json.	Status: live; priority: medium. Gate or predicate: validate outputs, validate manuscript. Negative control: missing fragment coverage. Scope: deterministic toy-only.
benchmark	Producer: generate toy sweep tracks.py. Current and next evidence: toy benchmark matrix.json.	Status: live; priority: medium. Gate or predicate: validate outputs. Negative control: missing fragment coverage. Scope: deterministic toy-only.
causal ablation	Producer: generate toy sweep tracks.py. Current and next evidence: causal ablation matrix.json.	Status: live; priority: high. Gate or predicate: validate outputs, validate manuscript. Negative control: causal ablation missing cell. Scope: deterministic toy-only.
counterexample	Producer: generate sheaf tracks.py. Current and next evidence: counterexample matrix.json.	Status: live; priority: high. Gate or predicate: validate outputs, validate manuscript. Negative control: known bad counterexample passed. Scope: deterministic toy-only.
evidence fields	Producer: generate sheaf tracks.py. Current and next evidence: evidence field index.json.	Status: live; priority: high. Gate or predicate: validate outputs, validate manuscript. Negative control: missing typed claim. Scope: deterministic toy-only.
formalism	Producer: compose manuscript.py. Current and next evidence: manifest.yaml.	Status: live; priority: medium. Gate or predicate: validate manuscript. Negative control: missing fragment coverage. Scope: deterministic toy-only.
gate ergonomics	Producer: generate integration audit.py. Current and next evidence: validation gate index.json.	Status: live; priority: high. Gate or predicate: validate outputs, validate manuscript. Negative control: gate ergonomics unindexed. Scope: deterministic toy-only.
gnn	Producer: generate formal interop tracks.py. Current and next evidence: gnn lint report.json.	Status: live; priority: medium. Gate or predicate: validate outputs. Negative control: missing fragment coverage. Scope: deterministic toy-only.
interop	Producer: generate formal interop tracks.py. Current and next evidence: interop roundtrip report.json.	Status: live; priority: high. Gate or predicate: validate outputs. Negative control: interop shape loss. Scope: deterministic toy-only.

Track	Evidence progression	Status, gate, and negative control
layers	Producer: generate figures.py. Current and next evidence: sheaf coverage matrix.json.	Status: optional; priority: medium. Gate or predicate: validate outputs, validate manuscript. Negative control: missing fragment coverage. Scope: deterministic toy-only.
lean	Producer: generate formal interop tracks.py. Current and next evidence: lean theorem inventory.json.	Status: live; priority: medium. Gate or predicate: validate outputs. Negative control: missing fragment coverage. Scope: deterministic toy-only.
manuscript staleness	Producer: z generate manuscript variables.py. Current and next evidence: manuscript staleness report.json.	Status: live; priority: medium. Gate or predicate: validate outputs, validate manuscript. Negative control: missing fragment coverage. Scope: deterministic toy-only.
model checking	Producer: generate sheaf tracks.py. Current and next evidence: model checking witnesses.json.	Status: live; priority: high. Gate or predicate: validate outputs. Negative control: missed model checking counterexample. Scope: deterministic toy-only.
ontology	Producer: generate formal interop tracks.py. Current and next evidence: ontology profile matrix.json.	Status: live; priority: medium. Gate or predicate: validate outputs. Negative control: missing fragment coverage. Scope: deterministic toy-only.
proof extraction	Producer: generate formal interop tracks.py. Current and next evidence: proof extraction index.json.	Status: live; priority: high. Gate or predicate: validate outputs, validate manuscript. Negative control: proof extraction missing statement. Scope: deterministic toy-only.
prose	Producer: compose manuscript.py. Current and next evidence: manifest.yaml.	Status: live; priority: medium. Gate or predicate: validate manuscript. Negative control: missing fragment coverage. Scope: deterministic toy-only.
provenance	Producer: generate sheaf tracks.py. Current and next evidence: artifact provenance.json.	Status: live; priority: high. Gate or predicate: validate manuscript, validate outputs. Negative control: missing sheaf track producer. Scope: deterministic toy-only.
pymdp	Producer: simulate si tmaze.py. Current and next evidence: si policy comparison.json.	Status: live; priority: medium. Gate or predicate: validate outputs. Negative control: missing fragment coverage. Scope: deterministic toy-only.
release bundle	Producer: generate sheaf tracks.py. Current and next evidence: release bundle manifest.json.	Status: live; priority: high. Gate or predicate: validate outputs, validate manuscript. Negative control: release bundle parity failure. Scope: deterministic toy-only.
release notes	Producer: generate integration audit.py. Current and next evidence: release notes evidence.json.	Status: live; priority: high. Gate or predicate: validate outputs, validate manuscript. Negative control: release notes claim failed gate passed. Scope: deterministic toy-only.
replay matrix	Producer: generate sheaf tracks.py. Current and next evidence: replay matrix.json.	Status: live; priority: high. Gate or predicate: validate outputs, validate manuscript. Negative control: replay mismatch. Scope: deterministic toy-only.
scholarship	Producer: generate sheaf tracks.py. Current and next evidence: scholarship source matrix.json.	Status: live; priority: high. Gate or predicate: validate outputs, validate manuscript. Negative control: missing scholarship source binding. Scope: deterministic toy-only.

Track	Evidence progression	Status, gate, and negative control
security posture	Producer: generate sheaf tracks.py. Current and next evidence: security posture audit.json.	Status: live; priority: high. Gate or predicate: validate outputs, validate manuscript. Negative control: security posture aggregate forgery. Scope: deterministic toy-only.
sensitivity	Producer: generate sheaf tracks.py. Current and next evidence: sensitivity sweep.json.	Status: live; priority: high. Gate or predicate: validate outputs. Negative control: missing sensitivity cell. Scope: deterministic toy-only.
simulation	Producer: generate toy sweep tracks.py. Current and next evidence: analytical observable sweep.json.	Status: live; priority: medium. Gate or predicate: validate outputs. Negative control: missing fragment coverage. Scope: deterministic toy-only.
state space catalog	Producer: generate toy sweep tracks.py. Current and next evidence: state space catalog.json.	Status: live; priority: high. Gate or predicate: validate outputs, validate manuscript. Negative control: state space catalog missing finite space. Scope: deterministic toy-only.
theorem traceability	Producer: generate sheaf tracks.py. Current and next evidence: theorem traceability matrix.json.	Status: live; priority: high. Gate or predicate: validate outputs, validate manuscript. Negative control: theorem traceability unlinked. Scope: deterministic toy-only.
uncertainty	Producer: generate sheaf tracks.py. Current and next evidence: uncertainty summary.json.	Status: live; priority: high. Gate or predicate: validate outputs. Negative control: unnormalized uncertainty row. Scope: deterministic toy-only.
visualization	Producer: generate integration audit.py. Current and next evidence: visualization quality audit.json.	Status: live; priority: medium. Gate or predicate: validate outputs, validate manuscript. Negative control: missing fragment coverage. Scope: deterministic toy-only.
empirical adapter	Producer: none while blocked. Current proof: blocked scope manifest.json. Next proving artifact: empirical adapter manifest.json.	Status: blocked; priority: blocked. Gate or predicate: blocked scope manifest all blocked. Negative control: empirical claim appears without manifest. Scope: blocked until provenance, licensing/privacy, deterministic replay, and typed claim gates exist.
private or restricted data	Producer: none while blocked. Current proof: blocked scope manifest.json. Next proving artifact: private data provenance manifest.json.	Status: blocked; priority: blocked. Gate or predicate: blocked scope manifest all blocked. Negative control: private data artifact appears without provenance manifest. Scope: blocked until provenance, licensing/privacy, deterministic replay, and typed claim gates exist.
network dependent research	Producer: none while blocked. Current proof: blocked scope manifest.json. Next proving artifact: network replay manifest.json.	Status: blocked; priority: blocked. Gate or predicate: blocked scope manifest all blocked. Negative control: network derived claim appears without replay manifest. Scope: blocked until provenance, licensing/privacy, deterministic replay, and typed claim gates exist.
llm generated evidence	Producer: none while blocked. Current proof: blocked scope manifest.json. Next proving artifact: llm evidence audit.json.	Status: blocked; priority: blocked. Gate or predicate: blocked scope manifest all blocked. Negative control: LLM generated evidence appears as a validation source. Scope: blocked until provenance, licensing/privacy, deterministic replay, and typed claim gates exist.

Track	Evidence progression	Status, gate, and negative control
non toy model claims	Producer: none while blocked. Current proof: blocked scope manifest.json. Next proving artifact: non toy model scope manifest.json.	Status: blocked; priority: blocked. Gate or predicate: blocked scope manifest all blocked. Negative control: non toy result claim appears outside future only scope. Scope: blocked until provenance, licensing/privacy, deterministic replay, and typed claim gates exist.

Improvement rows: 39 total; 32 live / 2 optional / 5 blocked; every row shown.

Results

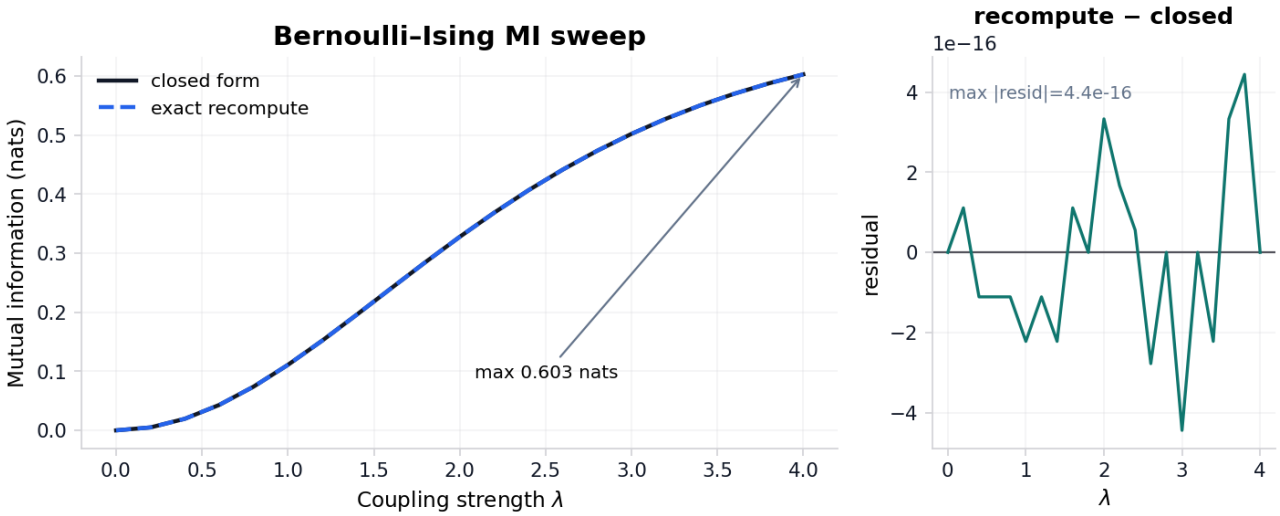
9 Mutual-information parameter sweep

We sweep coupling strength λ on a grid of 21 points up to $\lambda_{\max} = 4$. Closed-form mutual information from eq. 1 is cross-checked against an independent exact recomputation via total correlation from the analytical module (sec. 5); both are deterministic (no sampling) and agree to 0 nats.

Measured invariant checks: 12 / 12 passed on the clean tree.

The sweep reuses the entangled joint defined in eq. 1 (sec. 5). Mutual information $I(\lambda) = \log 2 - H_b(\sigma(\lambda))$ is evaluated on the same λ grid as the analytical oracle and its independent exact recomputation.

Both estimators are deterministic (no sampling, no RNG) and are evaluated on the same λ grid as the closed-form sweep (sec. 5, fig. 5).



Reproduced from fig. 5. Closed-form $I(\lambda)$ and an independent exact recomputation via total correlation for the symmetric Bernoulli-Ising toy across 21 grid points up to $\lambda_{\max} = 4$; grid maximum 0.6031 nats. Both estimators are deterministic (no sampling), so the right panel is a cross-implementation agreement check (max residual 0 nats), not a sampling residual.

10 Free-energy decomposition

Free energy against the entangled prior is evaluated along the same λ grid used for the MI sweep (fig. 15). Against the *entangled* prior the entangled posterior is the exact variational minimizer, so its free energy is identically zero; the Theorem-5.1 decomposition then splits that zero into per-stream marginal free energies, a coupling-cost term, a coupling-prior term, and a total-correlation gain. For the symmetric toy with uniform marginals the coupling-prior term equals $-I(\lambda)$ and exactly cancels the total-correlation gain $+I(\lambda)$ — an exact cancellation the merged invariant suite checks (12/12 pass). The curve in fig. 15 instead reports free energy against the *mean-field* prior: its minimum at $\lambda = 0$ is where the entangled posterior coincides with the factorized mean-field product, and any $\lambda > 0$ raises the free energy as coupling pulls the posterior away from that independent prior.

Saturation MI (grid maximum on the measured λ sweep): 0.6031 nats.

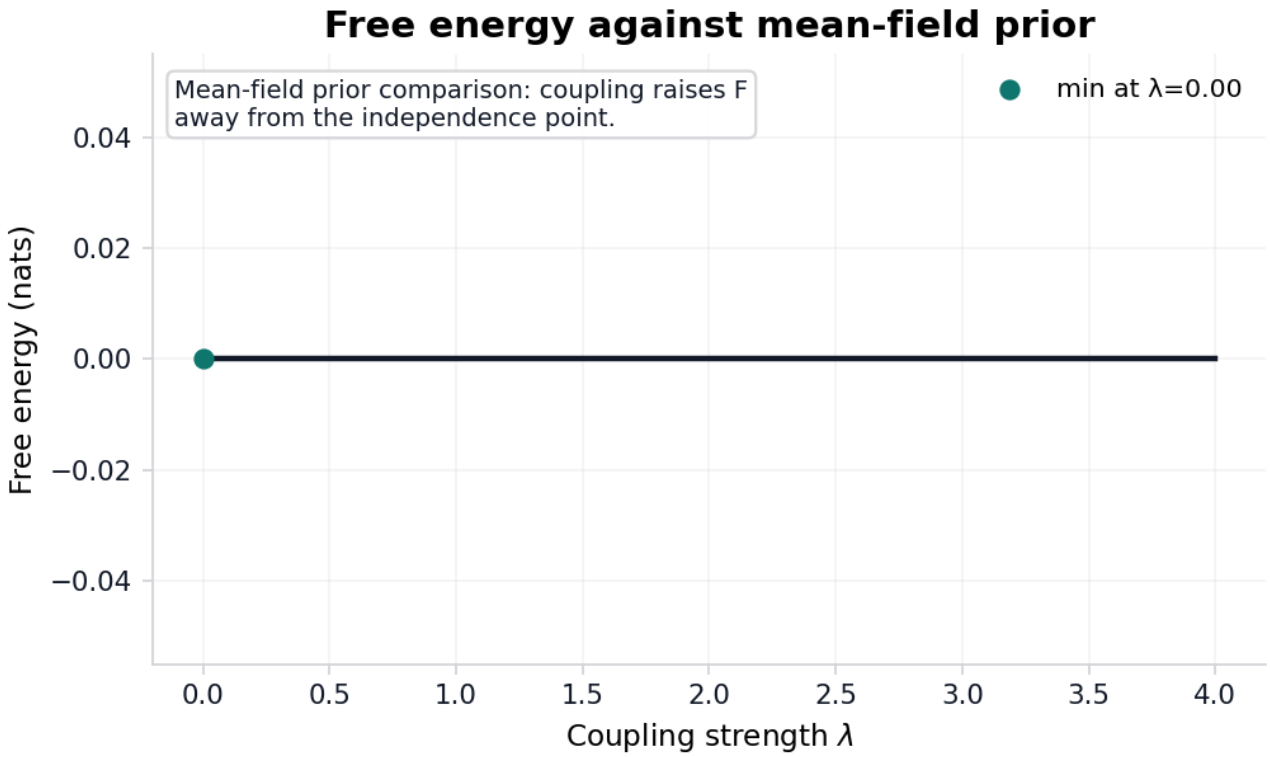


Figure 15: Free energy of the entangled posterior relative to the mean-field prior across the hyperparameter sweep (grid points 21); relative to the entangled prior, the same posterior has identically zero free energy.

11 T-maze active-inference rollout

The pymdp harness rolls out a T-maze active-inference agent in `state_inference` mode with planning horizon 2. The default `state_inference` mode is belief filtering with a goal-seeking action rule; sophisticated policy inference (an expected-free-energy policy posterior) is selectable via `mode: policy_inference` (sec. 6). Summary metrics land in `output/data/si_tmaze_summary.json`.

Steps recorded: 2. Mean belief entropy: 0. Belief entropy over the rollout is traced in fig. 20; the paired observation and action indices are in fig. 21. `output/data/analysis_statistics.json` now records the trace as a small statistical object rather than a caption-only trace: action switches 1 times (rate 1.000 over adjacent steps), observation diversity is 1, entropy drop is 0.0000 nats from first to terminal step, and the saved trace/summary step counts agree: `true` with finite entropy values `true`. The default `state_inference` mode runs pymdp `infer_states` and **reports** the resulting posterior (belief entropy and the state-1 marginal), but the action is chosen by an open-loop scripted rule on the observation index — not by the posterior — so the inferred belief here is observed, not acted on. Under the toy transition model, expected-free-energy policy inference reaches the goal in 1 of its rows versus 2 for the scripted state-inference rule: no behavioral advantage on this two-state, horizon-2 maze, which is the measured content of the deliberately-too-small claim.

Policy-comparison rows: 4 across state-inference and policy-inference modes; goal-reaching rows: 3. These rows are internal toy consistency checks under finite-horizon discrete active-inference assumptions [Friston et al., 2021, Da Costa et al., 2023], not comparisons against external behavioral datasets. Graph-world extension rows: 4 over 4 nodes, with goal-reached flag 1.

The expected free energy that scores those policies decomposes in closed form (fig. 17). Across the 4 length-2 policies on the T-maze generative model, the expected-free-energy-minimising policy is `00` with $G = 2.2539$ nats, splitting into risk 1.6037 (the pragmatic deviation of predicted outcomes from preferences) and ambiguity 0.6502 (the expected likelihood entropy) nats. The same G splits equivalently into pragmatic value -2.2539 (expected log-preference) and epistemic value 0.0000 (state-outcome mutual information) nats — the term that drives information-seeking. The two forms are exactly equal: risk + ambiguity + pragmatic + epistemic vanishes to within 0.0e+00 across every policy, the action-selection twin of the analytical free-energy decomposition identity (sec. 10).

Precision controls how sharply that expected free energy is acted on: the policy posterior is the softmax-weighted $q(\pi) \propto \exp(-\gamma\, G(\pi))$, and sweeping the inverse temperature γ across 33 grid points up to 16 sharpens it monotonically (fig. 18). Posterior entropy falls from $\ln 4$ at γ=0 to 1.1458 nats at γ=1, then saturates at the floor 0.6931 nats rather than reaching zero: the absorbing goal makes the second action irrelevant once reached, so 2 policies tie at the expected-free-energy minimum and precision concentrates mass on that optimal *set*, not a single policy. By that honest criterion selection becomes effectively deterministic — optimal-set mass exceeding 0.99 — at γ=3.

The minimal two-state maze above is, by construction, too small for information-seeking to matter: the reward location is observable from the start, so a greedy pragmatic rule and an expected-free-energy rule reach the goal equally often. The cue-then-reward variant (fig. 16) removes that degeneracy. Across 8 joint position-by-context states the reward location is an uninformative latent (50/50 at the start) that is hidden until the agent visits a CUE location, at which point a single sample resolves it: the cue carries 0.6931 nats of information ($= \ln 2$, the entropy of the unknown context). An agent that samples the cue and then takes the contingent arm reaches reward with expected log-preference -0.0538 nats, against -4.0538 nats for a greedy agent that commits to an arm before sampling — a measured behavioural advantage of 4.0000 nats that vanishes only if epistemic value is removed from the objective. The advantage is sophisticated, not flat: the closed-form flat decomposition of sec. 11 scores the cue-first and greedy policies as identical because it propagates beliefs through the transition model without conditioning them on the cue observation. Resolving the latent therefore requires an observation-conditioned (sophisticated-inference) evaluation, and under that evaluation cue-sampling is strictly necessary rather than merely available.

Planning is only half of the generative loop: the agent must also *learn* its likelihood. Placing a Dirichlet prior over each column of A and accumulating observation-state counts c gives the conjugate update $pA \leftarrow pA + c$ with expected likelihood $E[A] = pA / \sum_o pA$ (fig. 19). Driven by a fixed, sampling-free expected-count stream, $\mathrm{KL}(A_{\text{true}} \,\|\, A_{\text{learned}})$ falls monotonically from 0.7361 nats at the uniform prior to 1.29e-03 nats, reaching the convergence tolerance at update step 3. The learned likelihood converges to the true generative model in closed form, the inference-side twin of the EFE planning decomposition above.

Rollout trace: `output/data/si_tmaze_trace.json`. JSONL run log: `output/logs/pymdp_runs.jsonl`.

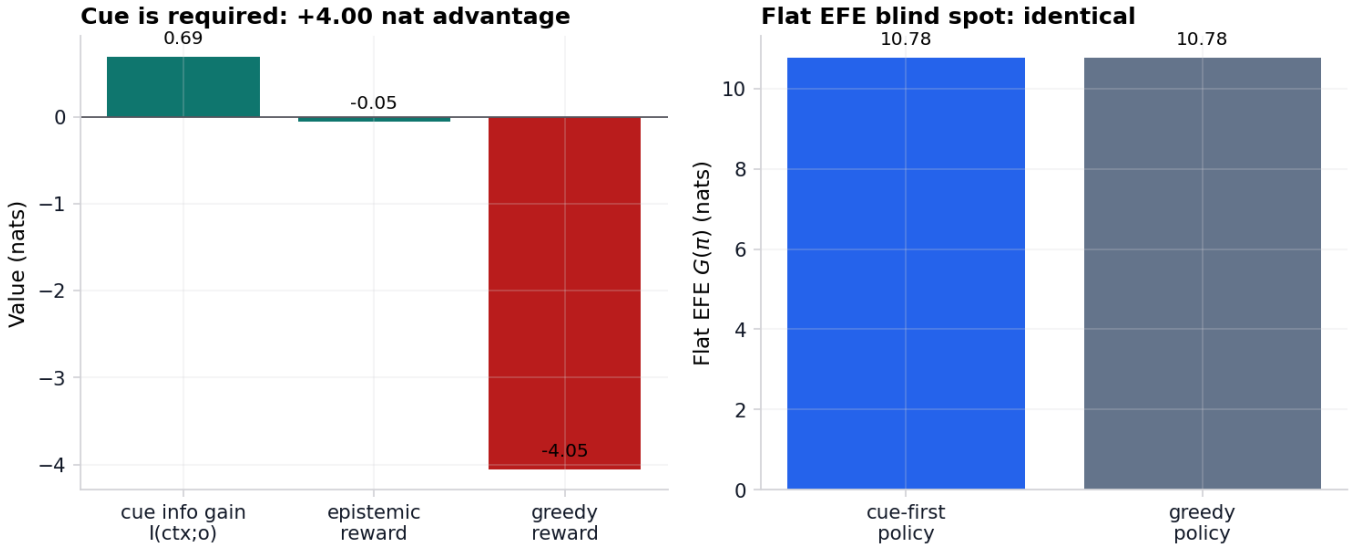


Figure 16: Cue-then-reward T-maze where epistemic value is strictly necessary. Left: the cue carries 0.6931 nats of information and the cue-sampling agent reaches reward with a measured advantage of 4.0000 nats over a greedy agent. Right: flat Expected Free Energy scores the cue-first and greedy policies identically (identical), so the sophisticated observation- conditioned evaluator is what makes information-seeking required. Closed form (no sampling).

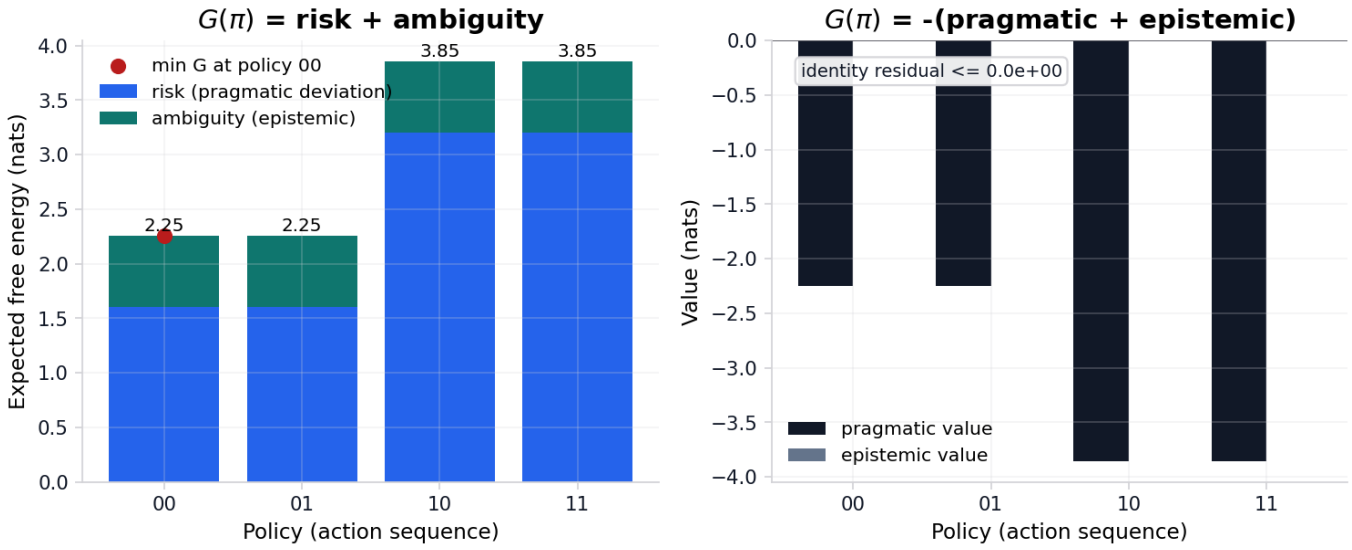


Figure 17: Closed-form Expected Free Energy decomposition over the finite T-maze policies. Left: $G(\pi)$ = risk + ambiguity (stacked), with the goal-seeking minimiser marked. Right: the pragmatic and epistemic values, which sum to $-G(\pi)$. Both forms are computed in closed form (no sampling) and satisfy risk + ambiguity + pragmatic + epistemic = 0 to machine precision.

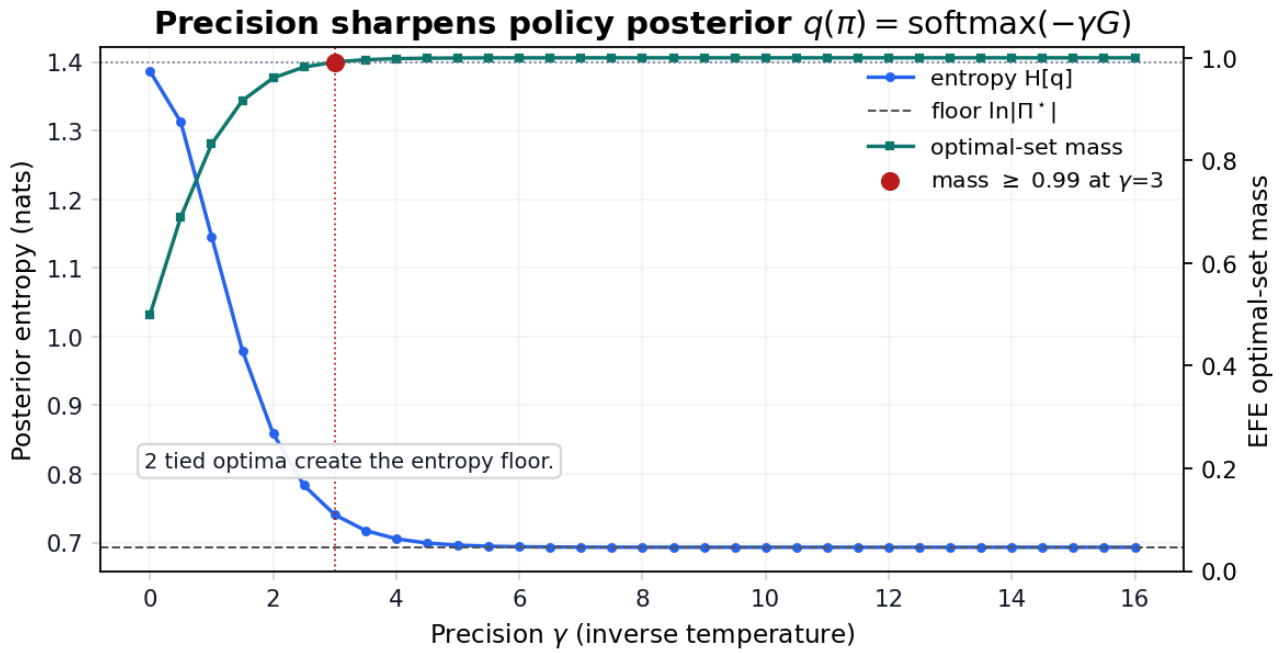


Figure 18: Precision sweep over the closed-form T-maze policy posterior $q(\pi) \propto \exp(-\gamma G)$ across 33 grid points up to γ=16. Entropy falls from $\ln 4$ to the $\ln|\Pi^\star|$ floor (2 tied optima, second action irrelevant under the absorbing goal); the optimal-set mass crosses 0.99 at γ=3. Entropy at γ=1 is 1.1458 nats. Computed in closed form (no sampling).

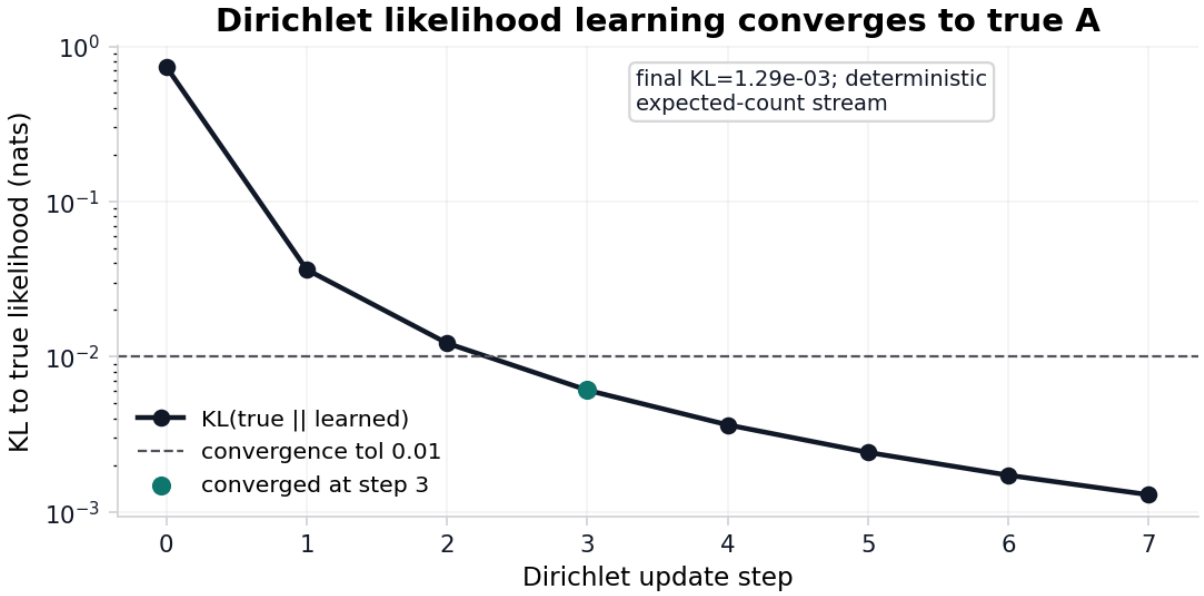


Figure 19: Dirichlet model learning: $\mathrm{KL}(A_{\text{true}} \,\|\, A_{\text{learned}})$ versus concentration-update step. The expected likelihood $E[A] = pA / \sum_o pA$ converges monotonically to the true generative likelihood; the run reaches the convergence tolerance at step 3 with final KL 1.29e-03 nats. Deterministic (no sampling), so the curve is byte-reproducible.

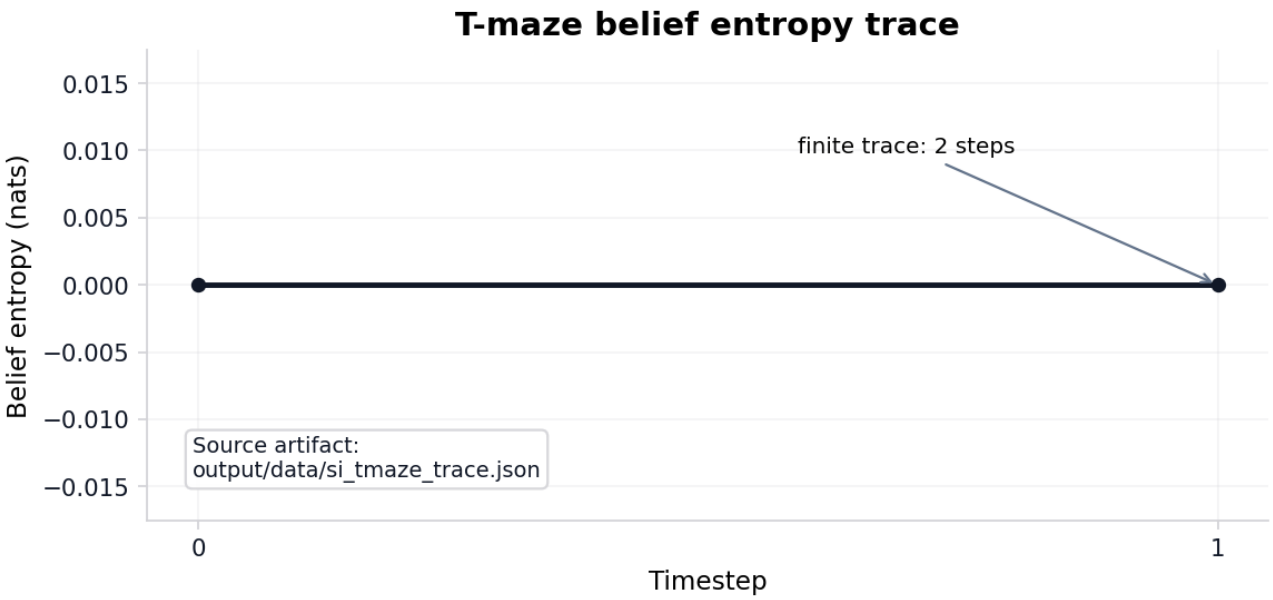


Figure 20: Belief entropy over time for the T-maze rollout (mean 0.0000 nats).

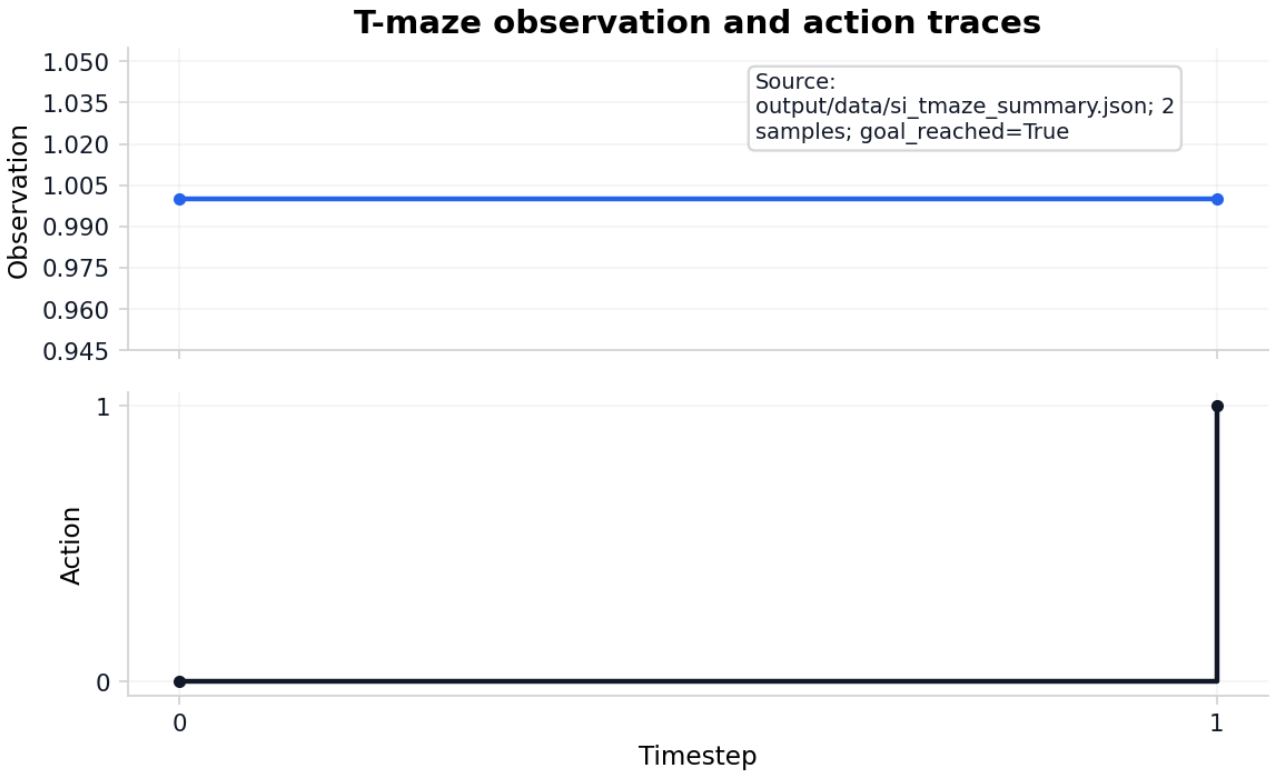


Figure 21: Observation and action traces for the T-maze rollout (action diversity 2).

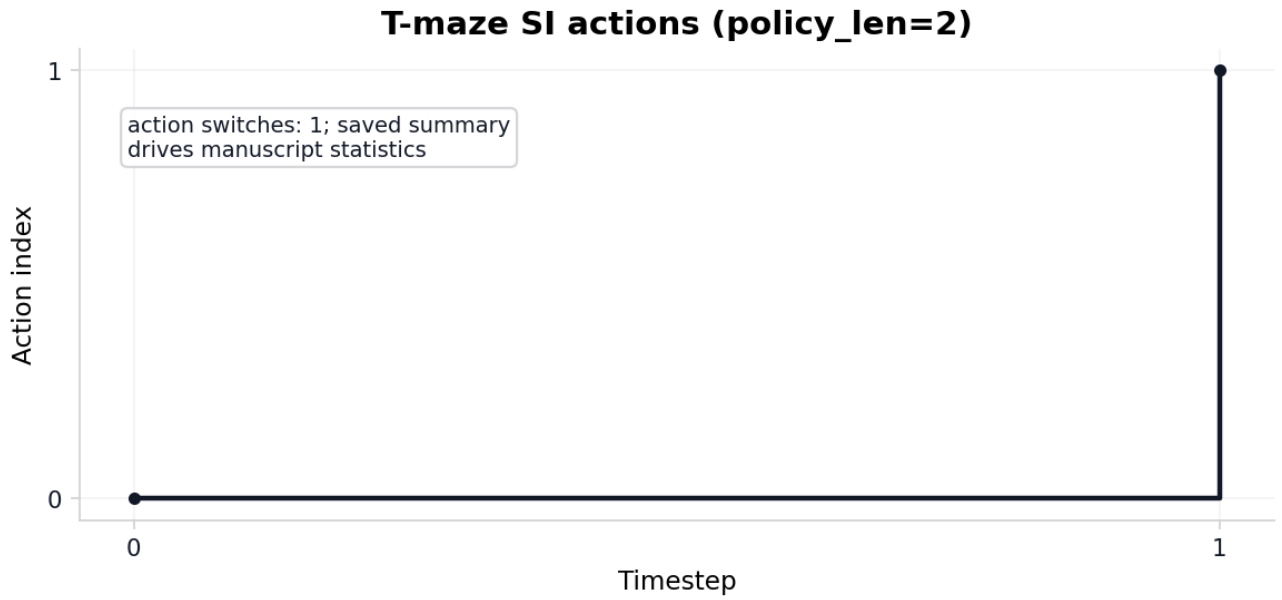


Figure 22: Discrete action index over time for the pymdp T-maze rollout (policy length 2).

12 Validation invariants

The analytical invariant registry runs before PDF rendering (sec. 5). On a clean checkout **12 / 12** checks pass in the merged validation report, which records simulation invariants when the pymdp harness ran (sec. 11).

fig. 23 lists each analytical and simulation gate; failures block publication artifacts. See sec. 8 for how invariant counts hydrate manuscript tokens.

Simulation invariants merge into the analytical report after the pymdp harness runs (sec. 11). fig. 23 summarizes pass/fail status for both domains on the clean tree.

The replay matrix exposes deterministic rerun comparison as table data rather than prose. It contains 13 producer rows, uses explicit replay-or-fingerprint methods, and every row must match its saved artifact hash (`true`).

The `sensitivity` fragment binds the deterministic toy sweep to the canonical sheaf track. `output/data/sensitivity_sweep.json` contains 96 cells across toy parameters, policy modes, seeds, horizons, and graph topologies; the hydrated flag `true` is the only manuscript claim about coverage.

The companion `output/data/si_policy_grid.json` records measured policy-mode rows derived from `si_policy_comparison.json`, not a synthetic grid. Missing cells fail the artifact schema before they can become prose; the topology trace artifact contributes 4 deterministic topology traces.

The `uncertainty` fragment reports only normalized toy summaries. `output/data/uncertainty_summary.json` contains 12 belief and policy-posterior rows plus 3 finite entropy bins, and `true` is false if any posterior row fails to sum to one within the deterministic tolerance.

Policy uncertainty is recorded in generated policy artifacts rather than hand-entered into the manuscript. The posterior grid contributes 5 available posterior rows; the EFE values artifact reports availability-or-measured-fallback flag 1. The fragment is therefore a validation surface, not an empirical uncertainty claim.

The `benchmark` fragment adds a compact toy matrix over the Bernoulli, T-maze, and graph-world artifacts. `output/data/toy_benchmark_matrix.json` reports 3 model rows and `true` only when each row names an artifact, metric, and passing gate.

The matrix is scoped to deterministic exemplar models. It is useful as a cross-track smoke test, not as a performance benchmark for biological or deployed systems.

12.0.1 State-space catalog track

The `state_space_catalog` track enumerates finite state spaces, action spaces, and policy counts for the deterministic toy models. The catalog artifact is `output/data/state_space_catalog.json`: it currently records 6 rows, with finite-space status `true`.

12.0.2 Causal ablation track

The `causal_ablation` track records deterministic toy ablations over finite preferences, likelihood-noise settings, and graph-topology perturbations. The matrix artifact is `output/data/causal_ablation_matrix.json`: it currently records 36 cells, with complete-grid status `true`.

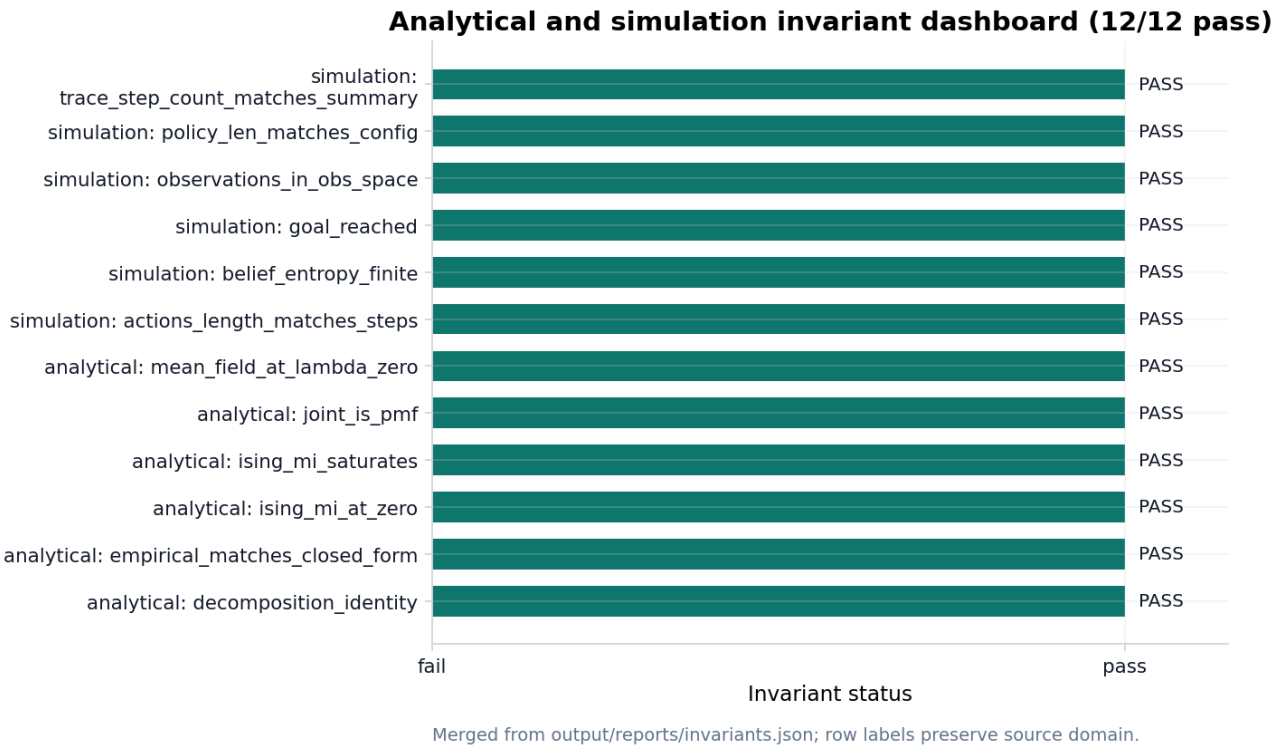


Figure 23: Invariant dashboard: 12 / 12 merged analytical and simulation checks from the validation registry.

Discussion

13 Limitations and outlook

13.1 What this demonstrates

The result of this manuscript is a *discipline*, not a domain claim: across three toy models every reported number is hydrated from a generated artifact, 6 sheaf axioms are machine-checked before composition, and 25 negative controls keep each failure path live. That posture follows the caution that FEP and active-inference formalisms need explicit methodological scope before they become empirical brain claims [Gershman, 2019]. No statistic, figure, or cross-track claim here can drift from its artifact without failing a gate before the PDF is built.

13.2 Limitations

The Bernoulli–Ising toy, T-maze harness, and sheaf composition model are pedagogical. They validate analytical consistency, artifact wiring, renderer dispatch, and manuscript hydration, not empirical claims about biological agents. Default pymdp mode is `state_inference` with planning horizon 2; the policy-comparison artifact exposes policy-inference rows without changing the default rollout (sec. 6).

13.3 Sheaf audit and outlook

sec. 1 and sec. 14 make binding state auditable under strict compose validation (sec. 8). Pipeline extensions in `tracks.yaml extension_tracks` now write deterministic artifacts: a belief GIF via `render_animation.py` and graph-world SI summary/trace via `simulate_si_graph_world.py`. The appendix row already binds an `animation` sheaf fragment without new manifest rows.

Sweep RMSE 0 nats and SI goal reached 1 summarize measured agreement on the declared grids and rollout. Future work includes full expected-free-energy policy selection, richer graph-world rollouts, and expanded Lean proofs beyond the boundary witnesses in sec. 7.

The discussion ontology binds `coverage_semantics` to the audit matrix in sec. 1, `pedagogical_scope` to the non-empirical scope of the toy models, and `state_inference_mode` to the pymdp harness contract in sec. 6.

Measured pymdp rollout (`state_inference`, config hash `133a13df72cfd8d8`): mean belief entropy 0.0000 nats over 2 steps; goal reached flag 1; action diversity 2.

Analytical sweep residual RMSE 0 nats (max residual 0). Coverage audit: 95 present / 95 bound / 0 missing cells on the IMRAD matrix.

The scholarship matrix is also a scope-control device. It separates conceptual lineage from measured evidence: cited sources explain why the toy models are relevant, while generated artifacts decide every numerical, figure, and gate claim. That split keeps the paper from converting background authority into an unsupported empirical result.

Sophisticated Learning is therefore cited as a future-only parameter-learning direction [Hodson et al., 2023]. It does not change the current T-maze state-inference default, does not promote an active-learning adapter, and does not alter the blocked major-scope ladder for empirical or non-toy claims.

13.3.1 Ontology bindings

- `coverage_semantics` → **Coverage matrix semantics**
- `pedagogical_scope` → **Pedagogical scope**
- `state_inference_mode` → **State inference harness**

13.3.2 Release notes evidence track

The `release_notes` track keeps release-language claims source-backed by validation, semantic, and bundle artifacts. Its evidence artifact is `output/reports/release_notes_evidence.json`: it currently records 3 rows, with source-backed status `true`.

Appendix

14 Appendix: full track coverage

This section is the **composability proof** for the manifest-indexed sheaf model: all 33 appendix-bound fragment tracks render into one flat manuscript section without section-specific compose branches. The registry defines 34 composable types; optional `layers` is methods-only and excluded from this row. The `animation` fragment is bound here as an optional registry type alongside the live proof, simulation, formal, notation, validation-spine, integration, audit, finite-catalog, ablation, license, release-evidence, scholarship, assumption-index, delta, and staleness tracks.

The proof is a publication-systems check (eq. 4). It demonstrates that heterogeneous fragments share one registry, manifest, renderer dispatch path, coverage matrix, and hydration boundary; it does not assert that every track carries equal scientific weight.

For each track $t \in \mathcal{T}_{\text{Full}}$, the appendix row binds a fragment path $f(t)$ and the composer emits `<!-- sheaf-track:t -->` before the rendered body. Generated renderers such as `section_figures` and markdown renderers pass through the same `resolve_track_body()` dispatch, so the appendix exercises the common compose interface rather than a bespoke appendix path.

$$|\mathcal{T}_{\text{Full}}| = 33 \tag{4}$$

The fragment registry defines 34 composable track types; optional `layers` is bound on the methods sheaf section only. Optional `animation` is bound in this appendix proof; the deterministic GIF artifact in `tracks.yaml extension_tracks` is produced by the core analysis DAG and remains separate from this fragment slot.

Because this appendix binds every non-optional appendix track plus optional `animation`, it is the maximal publication stalk of the coverage presheaf and exercises every publication renderer through the common `resolve_track_body()` dispatch. The same compose path is gated by the 6 sheaf laws verified in sec. 8 (6/6 satisfied): the appendix section glues to a unique output (separation), occupies the terminal position of the linear extension under its own `appendix` group row (poset and gluing), binds only well-typed fragments (typing), and owns every fragment path it references (compositionality). No count in this appendix is hand-written; all are injected from the registry-backed oracle.

Analytical sweep artifacts feed sec. 9 and sec. 12; simulation invariants merge after sec. 11. No additional path listing is required beyond those Results sections.

The appendix `assumption_index` row points to `output/data/analytical_assumption_index.json`. It binds 7 finite Bernoulli-Ising assumption rows to 7 equation identifiers and generated artifacts, with indexed status `true`.

The point is to make analytical signposting mechanical. If an equation is added without an assumption row, or if a row loses its evidence artifact, the index gate fails and the manuscript cannot present the equation as part of the validated finite toy proof surface.

pymdp harness summary: `output/data/si_tmaze_summary.json` (mean belief entropy, action trace). Runtime diagnostics: `output/reports/pymdp_runtime_diagnostics.json` (known warnings 4, unexpected warnings 0). Policy posterior grid: `output/data/pymdp_policy_posterior_grid.json` (10 rows). Full log: `output/logs/pymdp_runs.jsonl`.

`sheaf-track:interop` binds `output/data/interop_roundtrip_report.json`, `output/data/gnn_roundtrip_report.json`, `output/reports/gnn_lint_report.json`, and ontology profile artifacts into the appendix proof row. The appendix claim is exactly 2 checks with lossless status `true`.

The appendix provenance fragment points to `output/data/artifact_provenance.json`, the canonical artifact that records required toy artifact hashes, producer scripts, source commit, deterministic seeds, config digests, and 5 bundle rows.

`replay_matrix.json` provides the appendix proof for deterministic replay: 13 producer replay/fingerprint rows with matched status `true`.

The appendix counterexample fragment points to `output/reports/counterexample_matrix.json`, the expected-failure matrix that keeps promoted validation gates falsifiable. It currently records 25 known-bad fixtures, and the hydrated pass flag is `1`, meaning those fixtures are expected to fail rather than sneak through a positive-control gate.

This row is the negative-control ledger for the sheaf. Each counterexample names a promoted track, target validation gate, mutation, and observed expected-failure status. A new live track without a counterexample row is therefore visibly incomplete in the track-improvement scope.

`sheaf-track:adversarial_audit` binds `output/reports/adversarial_audit.json`, `output/reports/scope_boundary_audit.json`, and claim-audit outputs. The appendix claim is exactly 25 expected-failure rows with documented status `true` and known-bad-passing count 0.

`evidence_field_index.json` provides the appendix proof for field-level claim evidence: 97 mapped fields with status `true`.

`release_bundle_manifest.json` provides the appendix proof for required deliverables: 36 artifacts with source-present status `true`.

`artifact_contract_index.json` is the appendix-level cross-artifact concordance proof. It rederives 86 rows from the live semantic producer map and release-bundle parity rows, with row-complete flag `true` and copied-parity flag `true`.

`validation_gate_index.json` provides the appendix proof for gate ergonomics: 26 indexed gates. `track_lane_matrix.json` adds 32 pipeline-to-sheaf rows with completion flag `true`.

14.0.1 Appendix track: artifact diffoscope

`artifact_diffoscope` binds `output/reports/artifact_diffoscope.json` into the full sheaf appendix. Rows: 43. All equal: `true`.

This diffoscope is deliberately narrow and reproducibility-facing. For each non-cyclic generated artifact, it compares the saved provenance digest to the live file digest at validation time. The validator re-derives equality from the rows, so a stale `all_equal: true` summary cannot hide one changed artifact.

The row count is not a decoration; it is the number of artifact fingerprints that survived cycle exclusion and therefore can be compared directly. This keeps the release bundle honest about mutable files while avoiding self-referential hashes for artifacts that

necessarily include their own provenance.

14.0.2 Appendix track: artifact license

`artifact_license` binds `output/reports/artifact_license_audit.json` into the full sheaf appendix. Rows: 86. All safe: `true`.

The license audit classifies each generated or source-backed artifact under the public exemplar's configured license boundary. It is intentionally conservative: generated local outputs and project-owned source files pass, while an artifact outside those public source kinds would need an explicit provenance and license row before it could support a manuscript claim.

This is also where the blocked empirical-adapter boundary matters. Private, restricted, or network-derived data are not smuggled in as evidence; they remain blocked until privacy, licensing, typed-claim, semantic, and negative-control gates are implemented in the same artifact path.

`sheaf-track:scholarship` binds `output/data/scholarship_source_matrix.json` into the appendix proof row. The appendix claim is exactly 21 connected source rows with connected status `true`; each row names a bibliography key, locator, manuscript citation status, declared consumer sections, method role, registered track set, evidence artifact, and claim-boundary statement. The row set includes 3 quantitative/statistical or visualization-quality method roles, including 7 statistically backed figures with bridge status `true`. The explicit crosswalk has 7 rows and 9 statistical source links; every row is referenced in the manuscript (`true`), and every such reference section is manifest-bound to sheaf tracks (`true`) with a visualization track present (`true`). This binds statistics and figure-quality claims to generated artifacts rather than bibliography authority. The scholarship matrix itself also records manuscript citation presence (`true`), declared-section citation overlap count (19), scope-guarded boundaries (`true`), and live row re-derivation (`true`), which makes forged aggregate source-connectivity flags fail at the validation boundary.

The visualization registry is also now a paper-integration object: role metadata complete `true`, paper claims complete `true`, and section bindings complete `true`. Those flags are read from the saved visualization-quality audit and then rechecked through the semantic sheaf restrictions.

The appendix includes the security posture as a release-boundary proof object. Each row in `output/reports/security_posture_audit.json` has evidence artifacts, validators, a scoped boundary statement, and a negative-control identifier. The negative controls target the verifier failure modes a well-resourced adversary would prefer: aggregate forgery, untracked credentials, network-derived evidence, private-data leakage, unsigned production-release claims, and production zero-trust claims without a runtime identity plane.

The posture is therefore defensive and local: it hardens this public template against evidence laundering, artifact drift, secret exposure, and false release claims while keeping production-only controls deferred until a real deployment adds signed provenance, SBOMs, identity-aware access, telemetry, and incident response evidence.

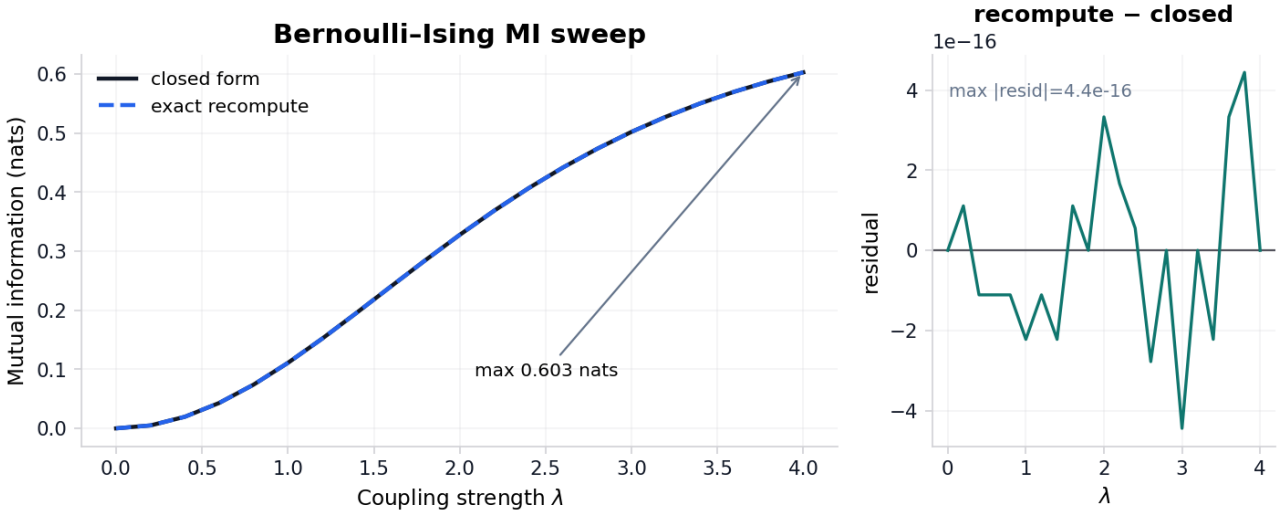
`sheaf-track:sensitivity` binds `output/data/sensitivity_sweep.json`, measured `output/data/si_policy_grid.json`, compatibility-named EFE values artifact `output/data/si_efe_terms.json`, `output/data/analytical_observable_sweep.json`, and graph-world topology artifacts including `output/data/si_graph_world_topology_traces.json`. The appendix claim is exactly 96 complete canonical grid cells.

`sheaf-track:uncertainty` binds `output/data/uncertainty_summary.json`. The appendix claim is exactly 12 normalized rows across 3 entropy bins with status `true`.

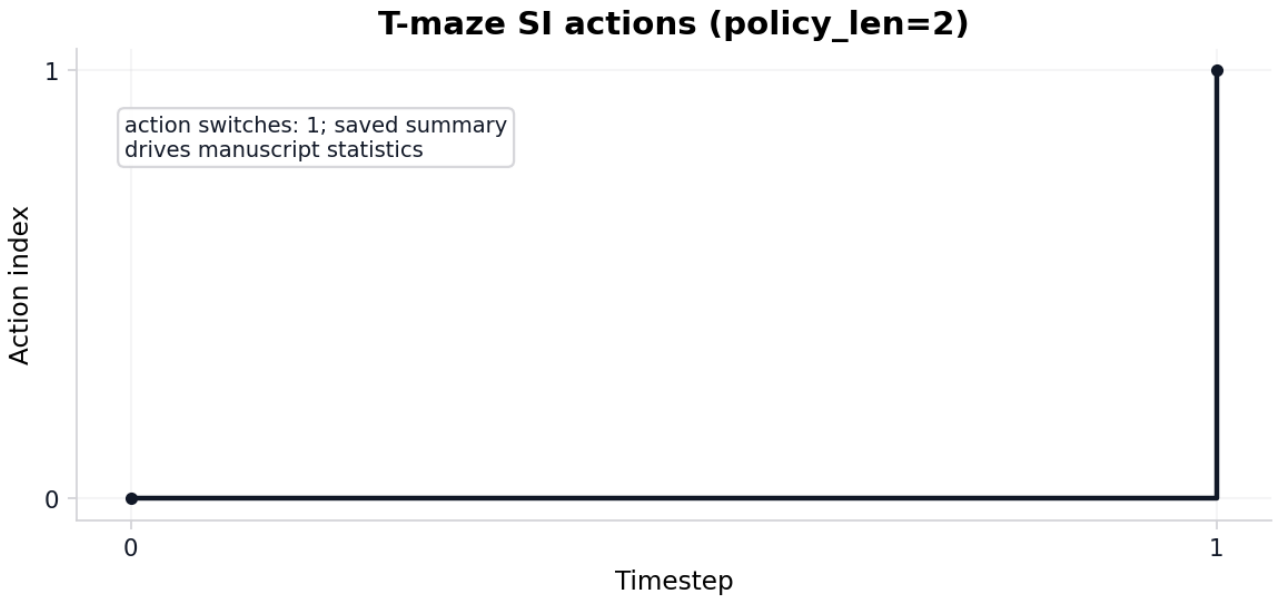
`sheaf-track:benchmark` binds `output/data/toy_benchmark_matrix.json`. The appendix claim is exactly 3 complete toy-model rows with status `true`.

The appendix `manuscript_staleness` row points to `output/reports/manuscript_staleness_report.json`. It checks 322 token bindings after hydration, including late audit variables, and the pass flag is `true`.

This is the rendered-output side of the sheaf contract. Source fragments may contain hydration placeholders, but the public manuscript must not; the staleness report compares each token's generated value against the resolved markdown so stale counts are caught after composition, not only during source-file linting.



Reproduced from fig. 5. Closed-form $I(\lambda)$ and an independent exact recomputation via total correlation for the symmetric Bernoulli-Ising toy across 21 grid points up to $\lambda_{\max} = 4$; grid maximum 0.6031 nats. Both estimators are deterministic (no sampling), so the right panel is a cross-implementation agreement check (max residual 0 nats), not a sampling residual.



Reproduced from fig. 22. Discrete action index over time for the pymdp T-maze rollout (policy length 2).

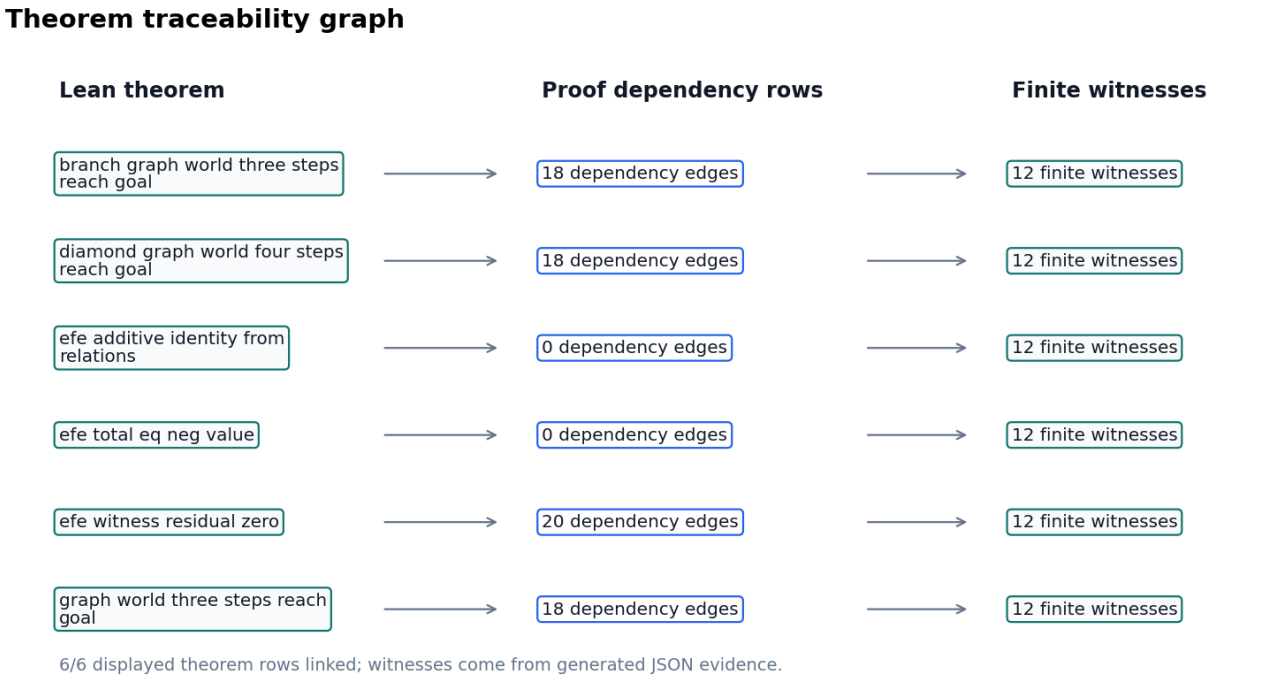
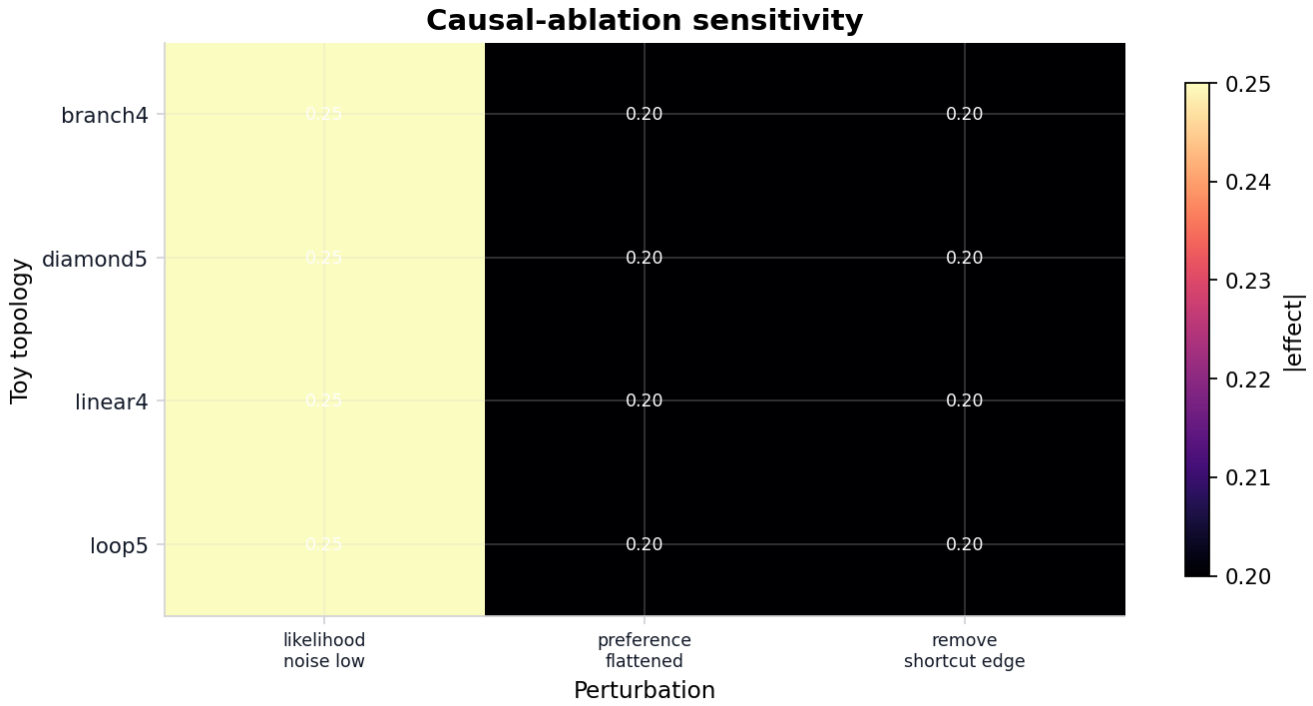


Figure 24: Theorem traceability graph generated from 17 linked theorem rows and 207 proof-dependency edges.



36 source rows; max absolute effect 0.250.

Figure 25: Causal-ablation heatmap: 36 source-backed rows joined to sensitivity and uncertainty artifacts; all effects source-backed: true.

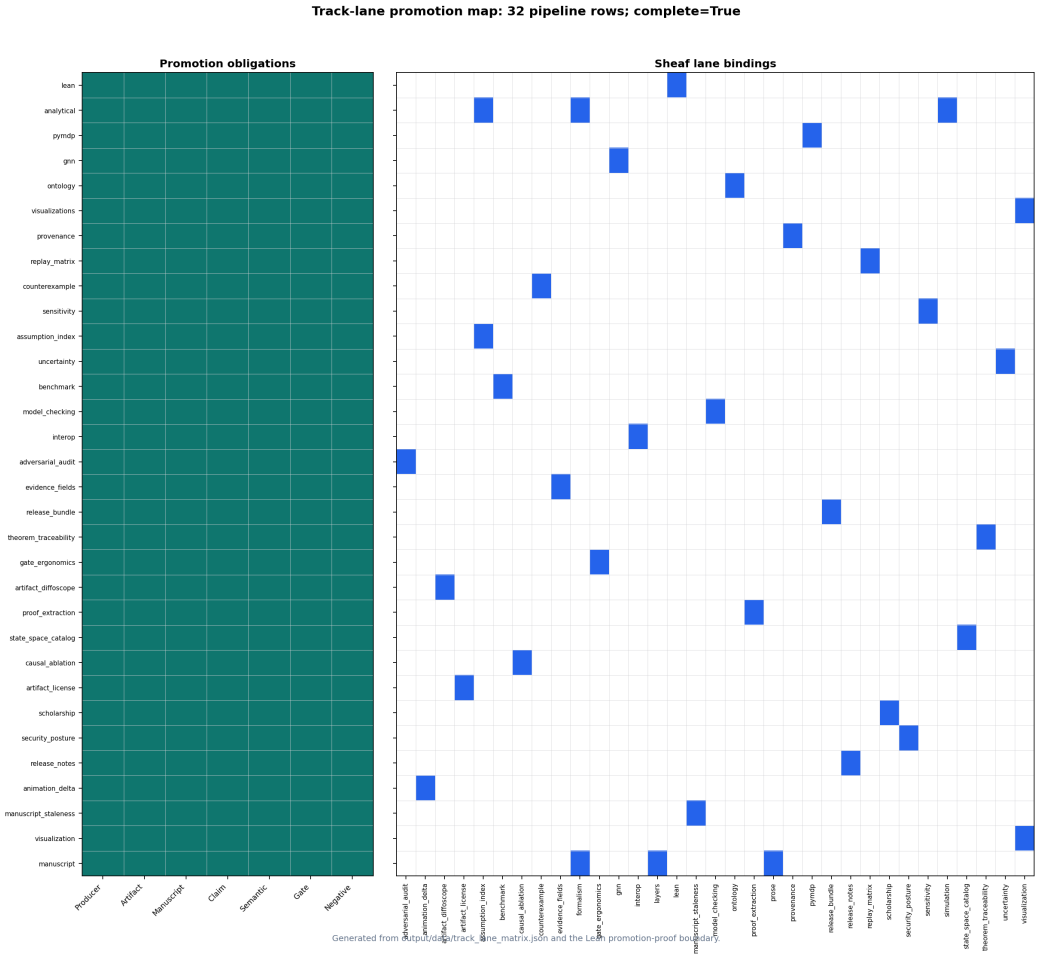
Scholarship source map with citation and scope guards

Citation	Locator	Scope boundary	Method role	Evidence artifact
@friston2010fep	doi+url cited in manuscript	guarded free energy principle	scope background	data/scholarship_source_matrix.json
@buckley2017mathreview	doi+url cited in manuscript	guarded free energy principle	mathematical review	data/analytical_assumption_index.json
@buckley2017mathreview	doi+url cited in manuscript	guarded free energy principle	statistics boundary	data/analysis_statistics.json
@dacosta2020discrete	doi+url cited in manuscript	guarded discrete state space active inference	discrete planning formalism	reports/model_checking_witnesses.json
@friston2021sophisticated	doi+url cited in manuscript	guarded sophisticated inference	sophisticated inference formalism	data/si_policy_grid.json
@dacosta2023reward	doi+url cited in manuscript	guarded discrete state space active inference	finite horizon reward planning	data/si_policy_comparison.json
@parr2022active	doi+url cited in manuscript	guarded active inference	conceptual reference	data/uncertainty_summary.json
@smith2022tutorial	doi+url cited in manuscript	guarded active inference practice	model building tutorial	data/si_policy_grid.json
@smith2022tutorial	doi+url cited in manuscript	guarded active inference practice	empirical scope boundary	reports/blocked_scope_manifest.json
@gershman2019fepbrain	doi+url cited in manuscript	guarded free energy principle	critical scope boundary	reports/blocked_scope_manifest.json
@pymdp2024	doi+url cited in manuscript	guarded pymdp implementation	implementation anchor	reports/pymdp_runtime_diagnostics.json
@champion2021branching	doi+url cited in manuscript	guarded sophisticated inference	branching time planning context	reports/model_checking_witnesses.json
@hodson2023sophisticatedlearning	doi+url cited in manuscript	guarded sophisticated inference	active learning future boundary	reports/blocked_scope_manifest.json
@nuijten2026efeplanning	doi+url cited in manuscript	guarded expected free energy planning	efe variational planning context	data/si_efe_terms.json
@gnn2023	doi+url cited in manuscript	guarded generalized notation notation	notation anchor	data/interop_roundtrip_report.json
@curry2014sheaves	doi+url cited in manuscript	guarded cellular sheaves	sheaf gluing background	data/sheaf_gluing_certificate.json
@curry2014sheaves	doi+url cited in manuscript	guarded cellular sheaves	finite sheaf law boundary	data/sheaf_section_status_matrix.json
@robinson2014topological	doi+url cited in manuscript	guarded applied sheaf signal processing	local to global consistency	data/sheaf_coverage_matrix.json
@robinson2014topological	doi+url cited in manuscript	guarded applied sheaf signal processing	visualization quality audit	reports/visualization_quality_audit.json
@bosca2026localglobal	doi+url cited in manuscript	guarded cellular sheaves	modern local to global context	data/sheaf_gluing_certificate.json
@buckley2017mathreview	doi+url cited in manuscript	guarded free energy principle	statistical visualization bridge	reports/visualization_quality_audit.json

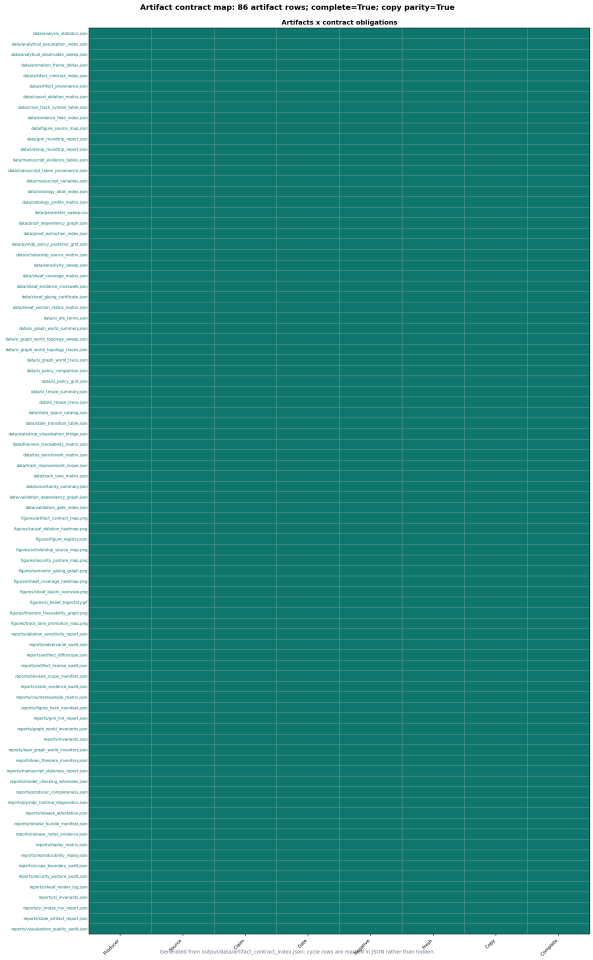
21 sources, 21 method roles, connected=True; rederived=True

Reproduced from fig. 13. Scholarship source map: 21 source rows across 21 method roles and 10 source families. Connected status:

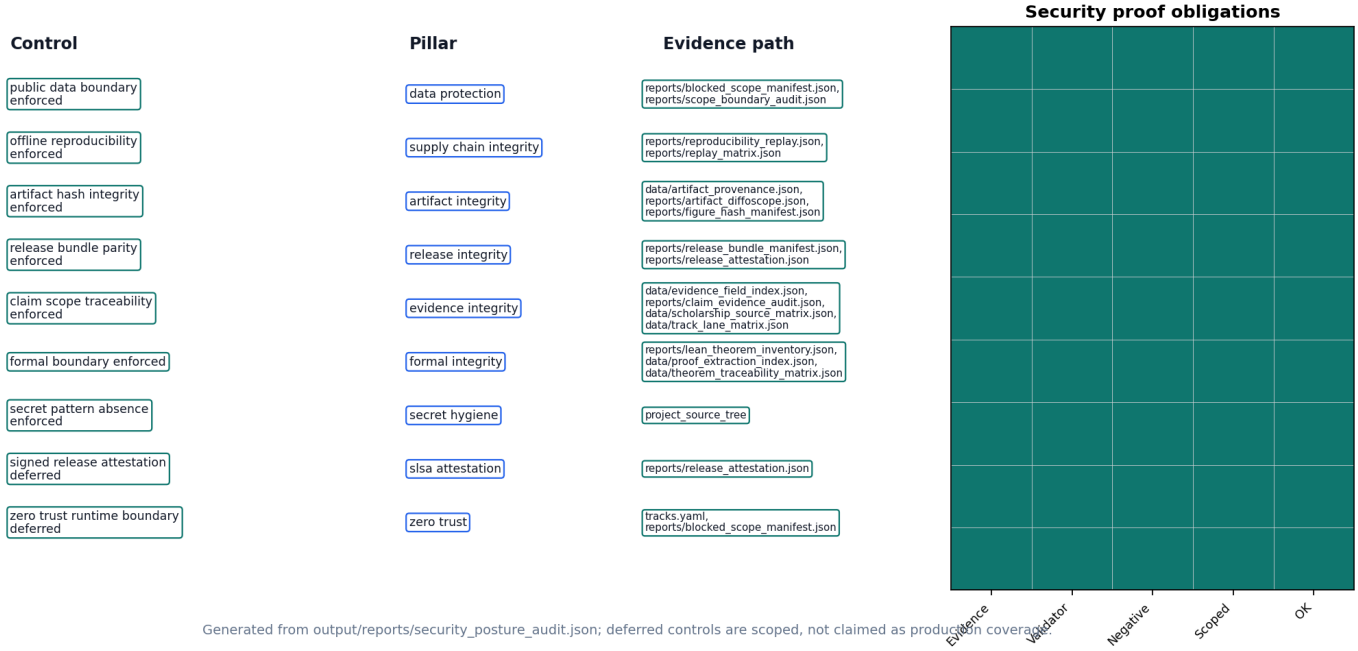
true; row evidence rederived: true.



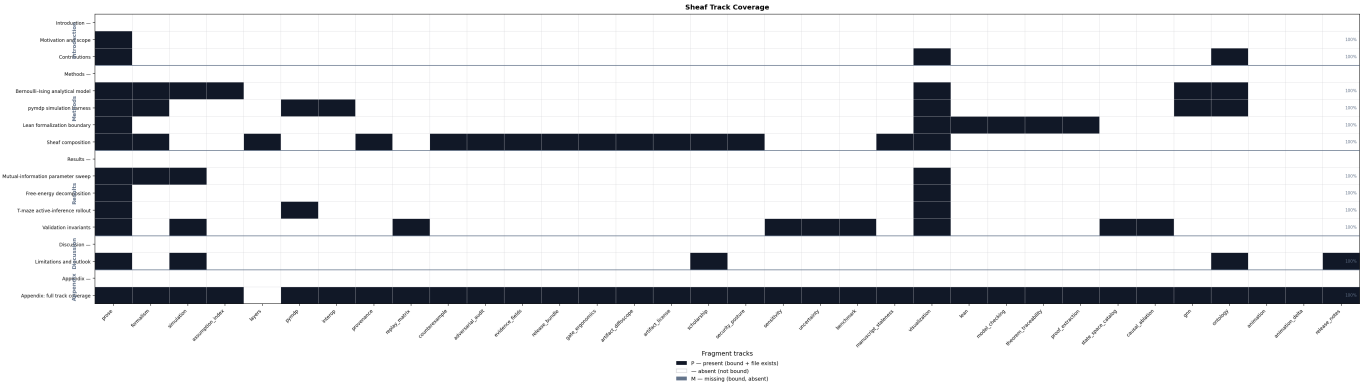
Reproduced from fig. 11. Track-lane promotion map: 32 pipeline-to-sheaf rows with complete promotion status true. Left: seven promotion-rule obligations; right: sheaf fragment bindings.



Reproduced from fig. 12. Artifact contract map: 86 generated artifact rows with complete contract status true and copied-output parity complete true. Cycle rows are explicit in `output/data/artifact_contract_index.json`.



Reproduced from fig. 14. Security posture map: 9 controls, 7 enforced and 2 scoped as deferred; secret findings: 0; high-risk gaps: 0.



Reproduced from fig. 3. Sheaf track coverage matrix: 17 IMRAD rows × 34 fragment columns. Black = present (P), white = absent (—), gray = missing (M). Counts: 95 present / 95 bound / 0 missing. Generated from `output/data/sheaf_coverage_matrix.json`.

Lean modules under `lean/TemplateActiveInference/` declare horizon and coupling witnesses. Build with `lake build` in `lean/`; fig. 8 summarizes proved versus deferred statements for this boundary fragment.

`sheaf-track:model_checking` binds `output/reports/model_checking_witnesses.json` and the Lean theorem inventories. The appendix claim is exactly 12 finite exhaustive witnesses with pass status `true`; Lean graph-world topology coverage is 4 generated topology ids with all-witnessed flag `true`.

`theorem_traceability_matrix.json` provides the appendix proof for theorem traceability: 17 linked rows with status `true`.

14.0.3 Appendix track: proof extraction

`proof_extraction` binds `output/data/proof_extraction_index.json` into the full sheaf appendix. Extracted theorems: 12. Constructive status: `true`.

The extraction index is intentionally modest: it records theorem names, statements, source files, leading tactics, and forbidden proof-token checks. That makes the Lean boundary inspectable without pretending that every proof term has been translated into a proof object. A row with a missing statement or forbidden token fails the formal interop gate and the canonical sheaf gate.

`output/data/proof_dependency_graph.json` adds the dependency view used by the appendix figure. It contributes 207 theorem-source, theorem-tactic, theorem-definition, and theorem-witness edges, with resolved edge status `true`; this is the artifact that keeps the theorem-traceability graph tied to generated Lean and model-checking rows.

14.0.4 Appendix track: state-space catalog

`state_space_catalog` binds `output/data/state_space_catalog.json` into the full sheaf appendix. Rows: 6. All finite: `true`.

The catalog is the finite-scope boundary for every toy claim in the exemplar. Each row records a model id, state count, action count, policy count, source artifact, and finite flag; the validator recomputes that counts are positive and that every row remains finite. This prevents a manuscript sentence about exhaustive checking from silently drifting into an unbounded or empirical setting.

`output/data/state_transition_table.json` makes the boundary operational. It contains 24 deterministic transition rows and covers all reachable finite models with status `true`. Readers can therefore audit not just the number of states, but the actual state/action/next-state relation used by the model-checking witnesses.

14.0.5 Appendix track: causal ablation

`causal_ablation` binds `output/data/causal_ablation_matrix.json` into the full sheaf appendix. Cells: 36. Complete grid: `true`.

The matrix is a finite teaching device: every row names a topology, a coupling value, a perturbation, a scalar effect, and the generated source row that made the effect admissible. It is not a claim about empirical interventions. It shows how an intervention-shaped table can be made falsifiable inside the sheaf: delete one perturbation cell or clear one deterministic flag and the grid gate fails before the manuscript can reuse the result.

`output/reports/ablation_sensitivity_report.json` then joins those ablation effects to the sensitivity and uncertainty artifacts. The report contributes 36 source-backed rows, with source-backed status `true`, so the appendix heatmap is a rendered view of validated JSON rather than a decorative restatement.

GNN declarations: `gnn/bernoulli_toy.gnn.md` and `gnn/si_tmaze.gnn.md` [Smékal and Friedman, 2023]. fig. 6 and sec. 5 document ontology concordance for the Bernoulli toy; SI notation extends the same pattern under sec. 6.

14.0.6 Ontology bindings

- `belief_entropy` → **BeliefEntropy**
- `expected_free_energy` → **ExpectedFreeEnergy**
- `location` → **HiddenState**
- `observation` → **ObservationLikelihood**
- `policy` → **PolicyPosterior**
- `sheaf_track` → **SheafFragment**

Animation is an **extension** sheaf track backed by a deterministic GIF from `scripts/render_animation.py`. This appendix row documents the track binding only; default publication still uses static SI figures (sec. 11, fig. 22) while the GIF remains an auditable

generated artifact.

The appendix `animation_delta` row points to `output/data/animation_frame_deltas.json`. The manifest records 3 adjacent-frame deltas, with `true` as the hydrated evidence that the GIF is trace-derived rather than a duplicated static frame.

14.0.7 Appendix track: release notes evidence

`release_notes` binds `output/reports/release_notes_evidence.json` into the full sheaf appendix. Rows: 3. Source-backed: `true`.

Release notes are treated as claims, not as informal changelog prose. Each row names a source artifact and a pass/deferred status, so the release note can say only what validation, bundle, or semantic artifacts support. The validator re-derives support from rows; flipping the summary bit without fixing a failed row still fails.

`output/reports/release_attestation.json` is the compact final view over the same boundary. It records 8 attestation rows for validation, release bundle hash, license audit, semantic certificate, and blocked-scope status, with all-attested flag `true`.

15 Conclusion

Analytical oracles (sec. 5), pymdp rollouts (sec. 11), and sheaf composition (sec. 8) share one auditable manuscript contract: measured artifacts hydrate 12 composed sections, sec. 1 reports binding state, and strict compose validation blocks gray matrix cells before PDF rendering.

The T-maze harness runs in `state_inference` mode with config hash `133a13df72cfd8d8`; sweep RMSE 0 nats summarizes analytical-empirical agreement on the toy coupling grid. sec. 12 merges analytical and simulation gates; sec. 13 states scope and extensions. Scientific claims remain confined to declared models, not empirical statements about biological agents [Gershman, 2019].

The exemplar therefore sits at a narrow intersection: finite discrete active inference on POMDP-like toy models [Da Costa et al., 2020, Friston et al., 2021, Da Costa et al., 2023], sheaf-style local-to-global checks for publication artifacts [Curry, 2014, Robinson, 2014], and GNN notation as an interop layer [Smékal and Friedman, 2023].

16 References

See `manuscript/references.bib` for bibliography entries cited in the composed sections.

Vicente Bosca and Robert Ghrist. Neural networks as local-to-global computations, 2026. URL https://arxiv.org/abs/2603.14831.

Christopher L. Buckley, Chang Sub Kim, Simon McGregor, and Anil K. Seth. The free energy principle for action and perception: A mathematical review. *Journal of Mathematical Psychology*, 81:55–79, 2017. doi: 10.1016/j.jmp.2017.09.004. URL https://doi.org/10.1016/j.jmp.2017.09.004.

Théophile Champion, Lancelot Da Costa, Howard Bowman, and Marek Grześ. Branching time active inference: The theory and its generality, 2021. URL https://arxiv.org/abs/2111.11107.

Justin Michael Curry. *Sheaves, Cosheaves and Applications*. PhD thesis, University of Pennsylvania, 2014. URL https://repository.upenn.edu/entities/publication/c391a10c-f2e5-40be-bc3f-e6f73a43ddfb.

Lancelot Da Costa, Thomas Parr, Noor Sajid, Sebastijan Veselic, Victorita Neacsu, and Karl Friston. Active inference on discrete state-spaces: A synthesis. *Journal of Mathematical Psychology*, 99:102447, 2020. doi: 10.1016/j.jmp.2020.102447. URL https://doi.org/10.1016/j.jmp.2020.102447.

Lancelot Da Costa, Noor Sajid, Thomas Parr, Karl Friston, and Ryan Smith. Reward maximization through discrete active inference. *Neural Computation*, 35(5):807–852, 2023. doi: 10.1162/neco_a_01574. URL https://doi.org/10.1162/neco_a_01574.

Karl Friston. The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138, 2010. doi: 10.1038/nrn2787. URL https://www.nature.com/articles/nrn2787.

Karl Friston, Lancelot Da Costa, Danijar Hafner, Casper Hesp, and Thomas Parr. Sophisticated inference. *Neural Computation*, 33 (3):713–763, 2021. doi: 10.1162/neco_a_01351. URL https://doi.org/10.1162/neco_a_01351.

Samuel J. Gershman. What does the free energy principle tell us about the brain? *Neurons, Behavior, Data Analysis, and Theory*, 2 (3):1–10, 2019. doi: 10.51628/001c.10839. URL https://doi.org/10.51628/001c.10839.

Conor Heins, Beren Millidge, Daphne Demekas, Brennan Klein, Karl Friston, Iain D. Couzin, and Alexander Tschantz. pymdp: A python library for active inference in discrete state spaces. *Journal of Open Source Software*, 7(73):4098, 2022. doi: 10.21105/joss.04098. URL https://doi.org/10.21105/joss.04098.

Rowan Hodson, Bruce Bassett, Charel van Hoof, Benjamin Rosman, Mark Solms, Jonathan P. Shock, and Ryan Smith. Sophisticated learning: A novel algorithm for active learning during model-based planning, 2023. URL https://arxiv.org/abs/2308.08029.

Wouter W. L. Nuijten, Thijs van de Laar, and Bert de Vries. Expected free energy-based planning as variational inference, 2026. URL https://arxiv.org/abs/2606.20658.

Thomas Parr, Giovanni Pezzulo, and Karl J. Friston. *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press, 2022. doi: 10.7551/mitpress/12441.001.0001. URL https://mitpress.mit.edu/9780262045353/active-inference/.

Michael Robinson. *Topological Signal Processing*. Springer, 2014. doi: 10.1007/978-3-642-36104-3. URL https://link.springer.com/book/10.1007/978-3-642-36104-3.

Jakub Smékal and Daniel Ari Friedman. Generalized notation notation for active inference models, 2023. URL https://zenodo.org/records/7803328.

Ryan Smith, Karl J. Friston, and Christopher J. Whyte. A step-by-step tutorial on active inference and its application to empirical data. *Journal of Mathematical Psychology*, 107:102632, 2022. doi: 10.1016/j.jmp.2021.102632. URL https://doi.org/10.1016/j.jmp.2021.102632.

END OF TRANSMISSION

Release: v0.4.0 · DOI 10.5281/zenodo.20417021 · SHA-256 pending… · pairing pending



Figure 26: Integrity QR strip

Prior: v0.3.2 · 10.5281/zenodo.20931870 · f191b48f…